

Recap before meeting 26-05-2025



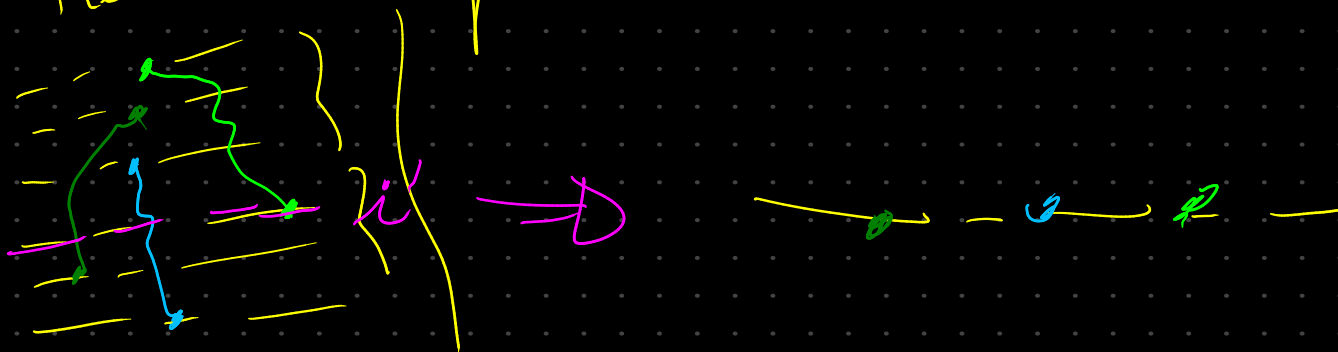
$$\forall i \in L(h) \quad \exists i' \in L(h^*) \quad \forall v \in V(h)$$

s.t. $\ell(v) = i \Rightarrow$ its $e(v)$ crosses

level i' in h^*

i' is part of the representation of i in h^*

we take the order in which the $e(v)$ $\forall v \in V(h)$ with $\ell(v) = i$ and determine the order of v_s in h .



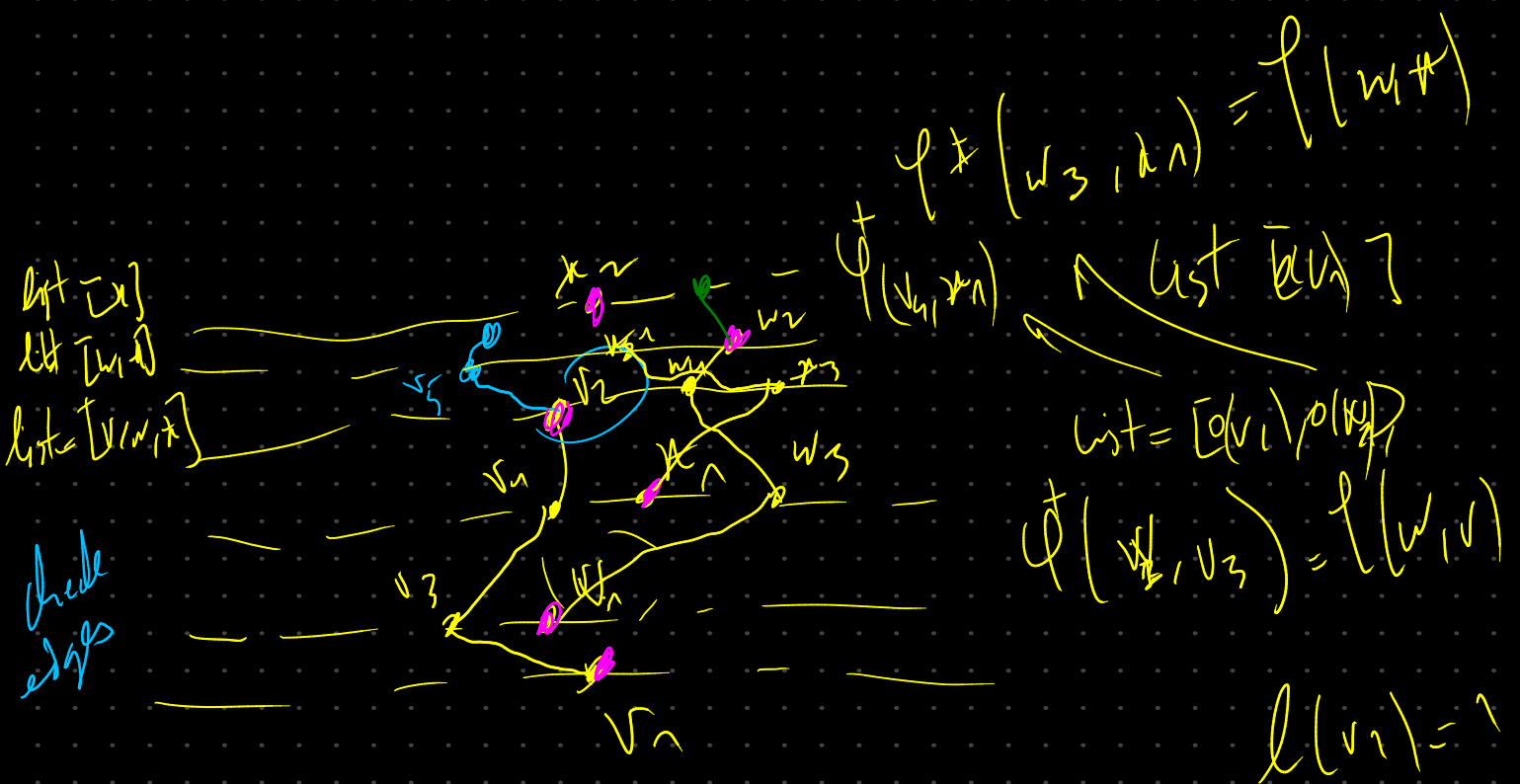
So we can draw a level planar drawing of G from a level planar drawing of G^*

we go line by line, the first level that we find all our stretch edge intersecting in it, we determine the order of the vertices corresponding to the mapped level in G .

$$(R^* \rightarrow R \quad \checkmark)$$

step 1: $S(G) \rightarrow S(G^*)$

$$\forall i \in L(G) \quad \forall v \in V(G) \text{ with } l(v) = i$$

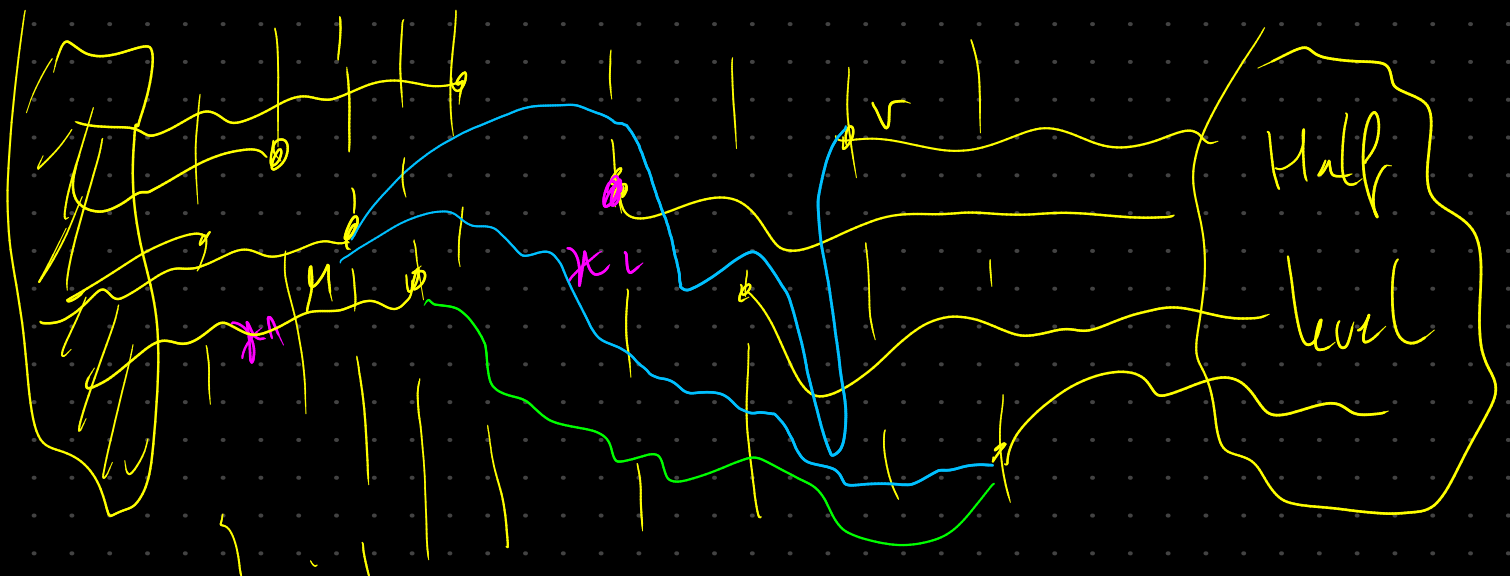


$S(L) \rightarrow S(L^+) \rightarrow \mathcal{P}^*$ of L^* (doesn't change any relation order)

$\mathcal{P}^*$ of L^* (H-T) $\rightarrow$ weak HT $\rightarrow$ λ -monotone L^*

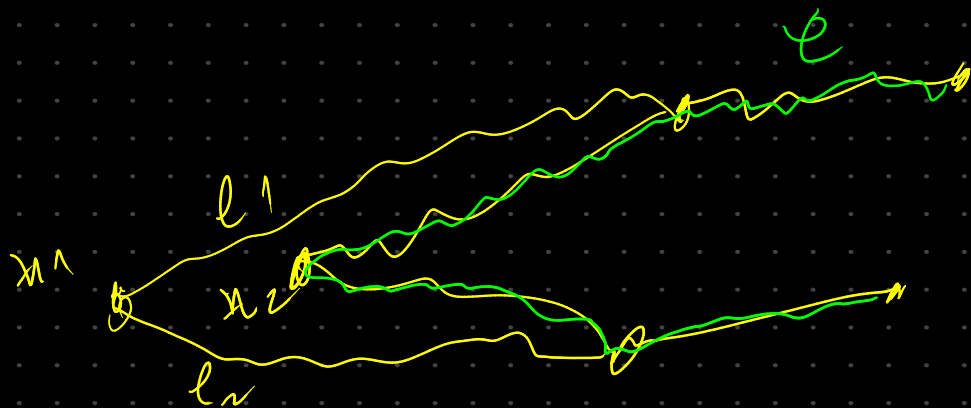
$\mathcal{P}^*$ is composed of L -components between levels corresponding to the same level of L .

but then there is an edge (u, v) with $L_u(u) \neq L_u(v)$, in this case we find ourselves in



case of L -in multiple comp.

the only time we can have this case, is when x_2 doesn't have any edge on the left.
 in this case we will always end up with multiple connected comps.



$h - \{x_1\} \rightarrow 1 \text{ comp}$

we can't have the case of $h - \{x_1\}$ multiple comps with



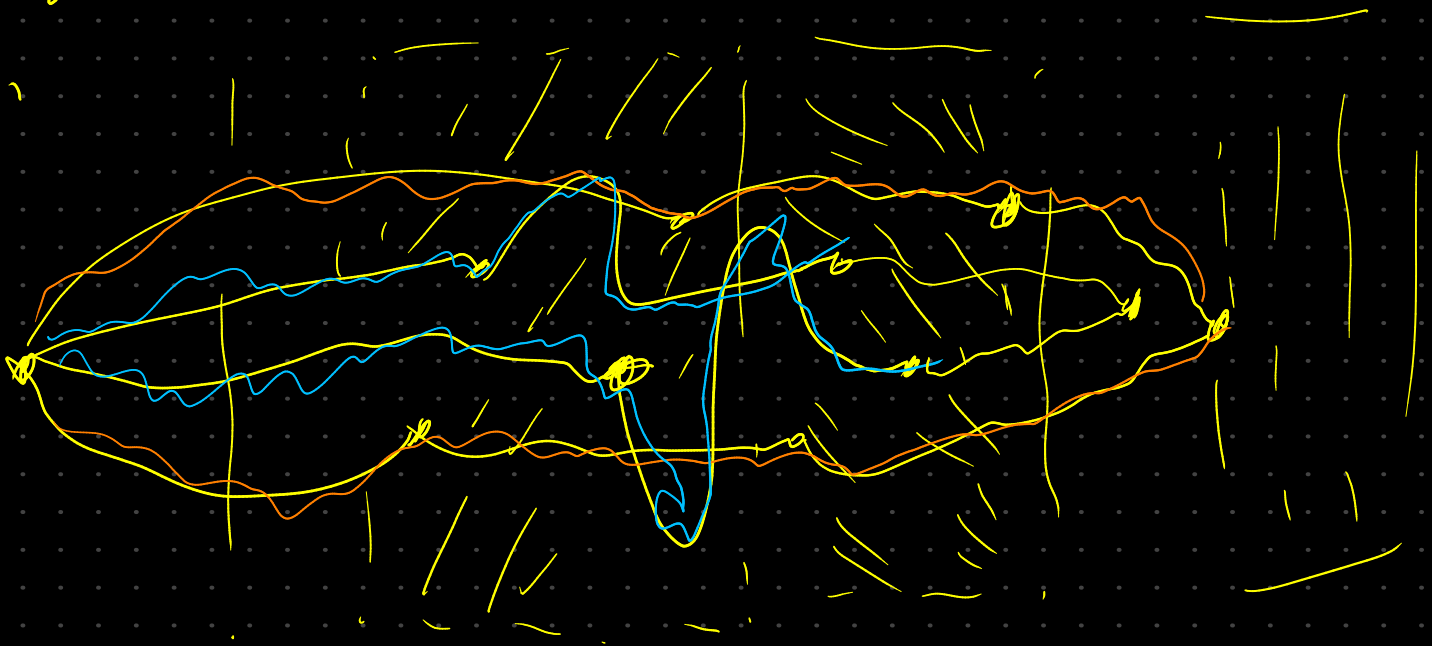
because the stretch edges are not connected to each other, so we can't have a H with

$$\begin{cases} e_1 > e_2 \\ e_1 > f_2 \end{cases} \quad \& \quad \begin{cases} f_1 < e_2 \\ f_1 < 1 \end{cases}$$

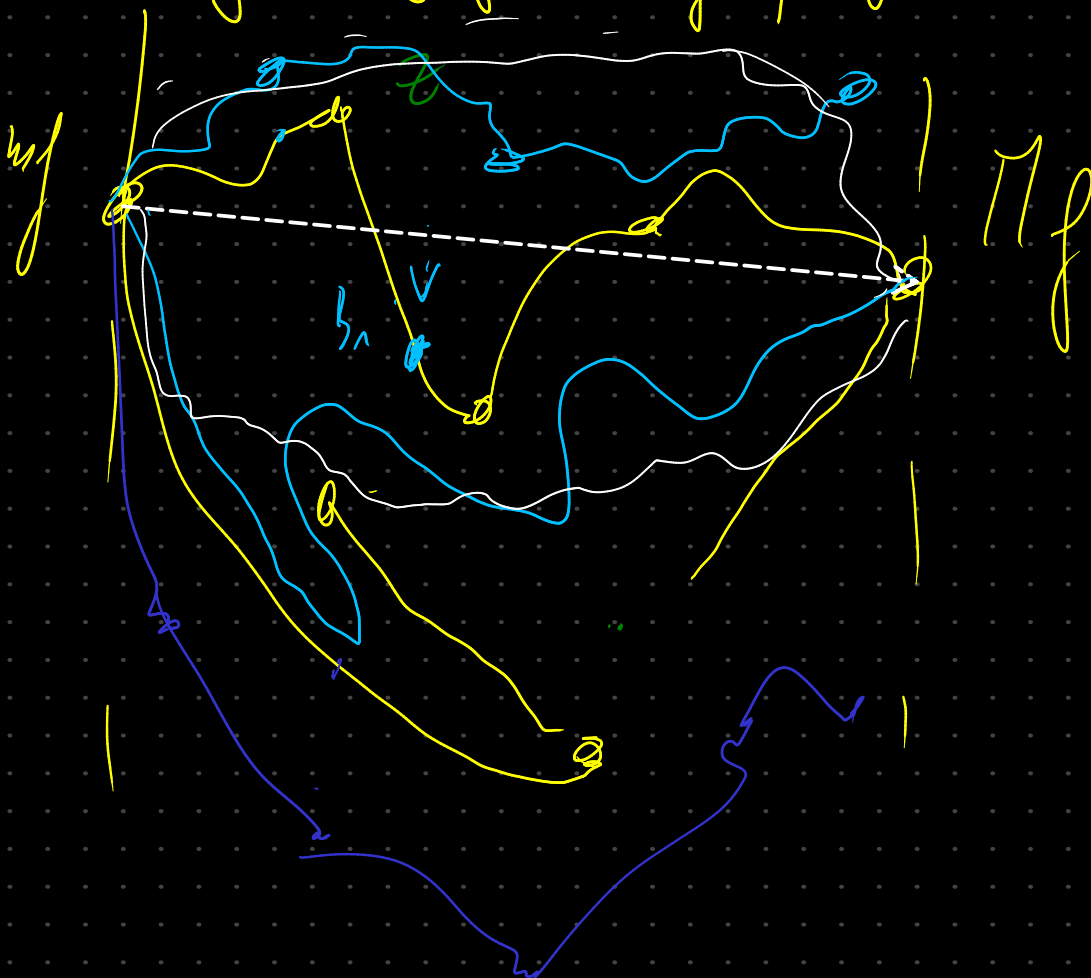
→ we're not in the case that we lemma

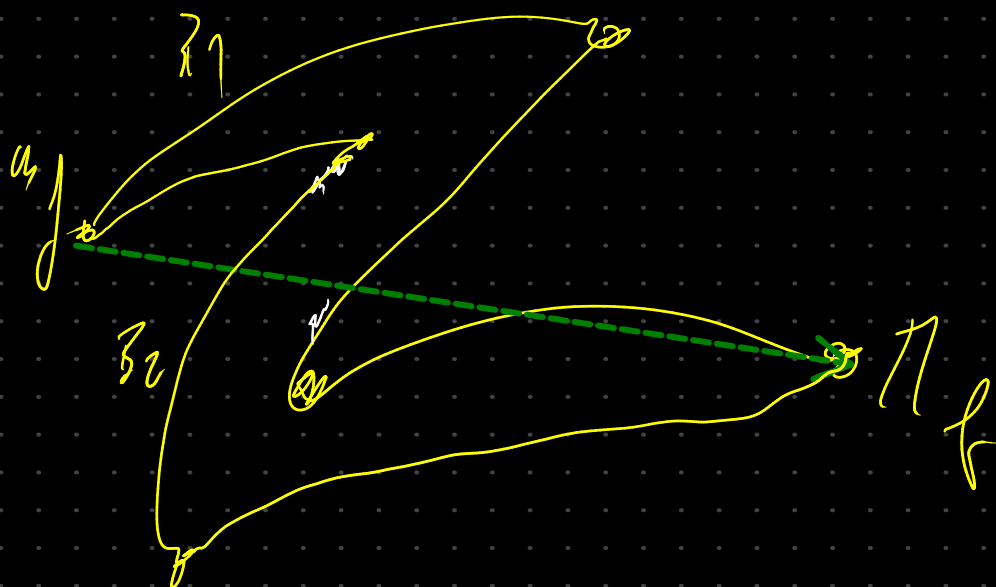
2.

3.



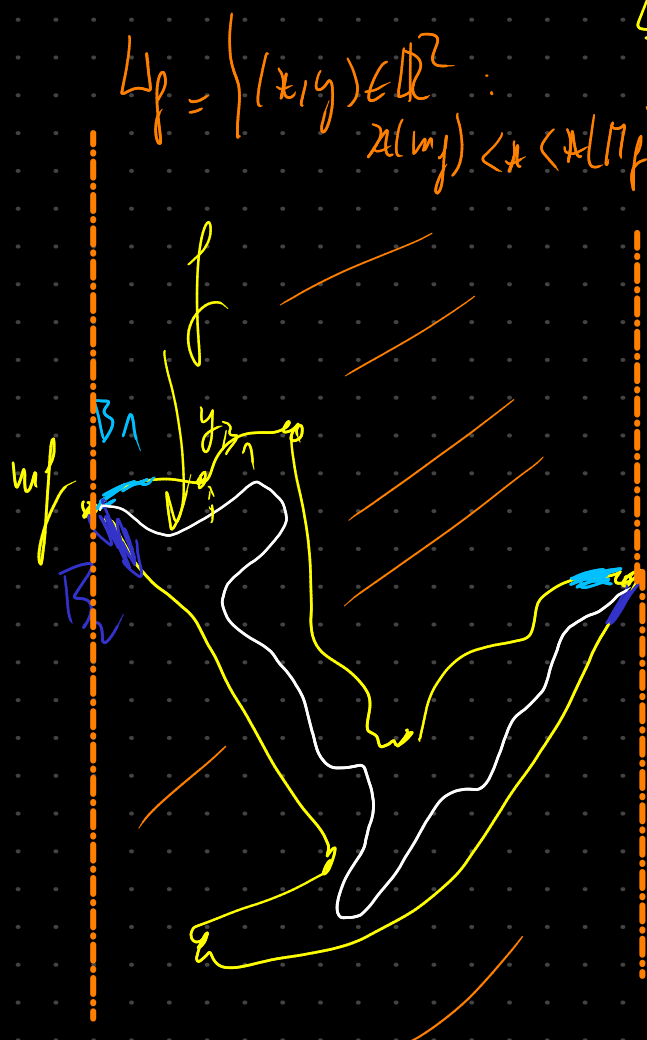
Pf drawing of $L \cap L_f := \{ (x, y) \in \mathbb{R}^2 : x(l_f) < x < x(l_f) \}$





We assume h is convex. Then every face
is bounded by a closed walk (why every f ?)

Because adding (u_f, π_f) will lead to a closed walk
(still not sure about "every")



π_f Redraw h locally
in U_f

goal: (Draw $m_f H_f$ as a straight-line seg)



$\forall z \in D_f$ (or) $\downarrow$

$$\forall x \in [x(m_f), x(\Pi_f)]$$

$$\exists y \in \mathbb{R} \text{ s.t. } (x, y) \in B_1$$

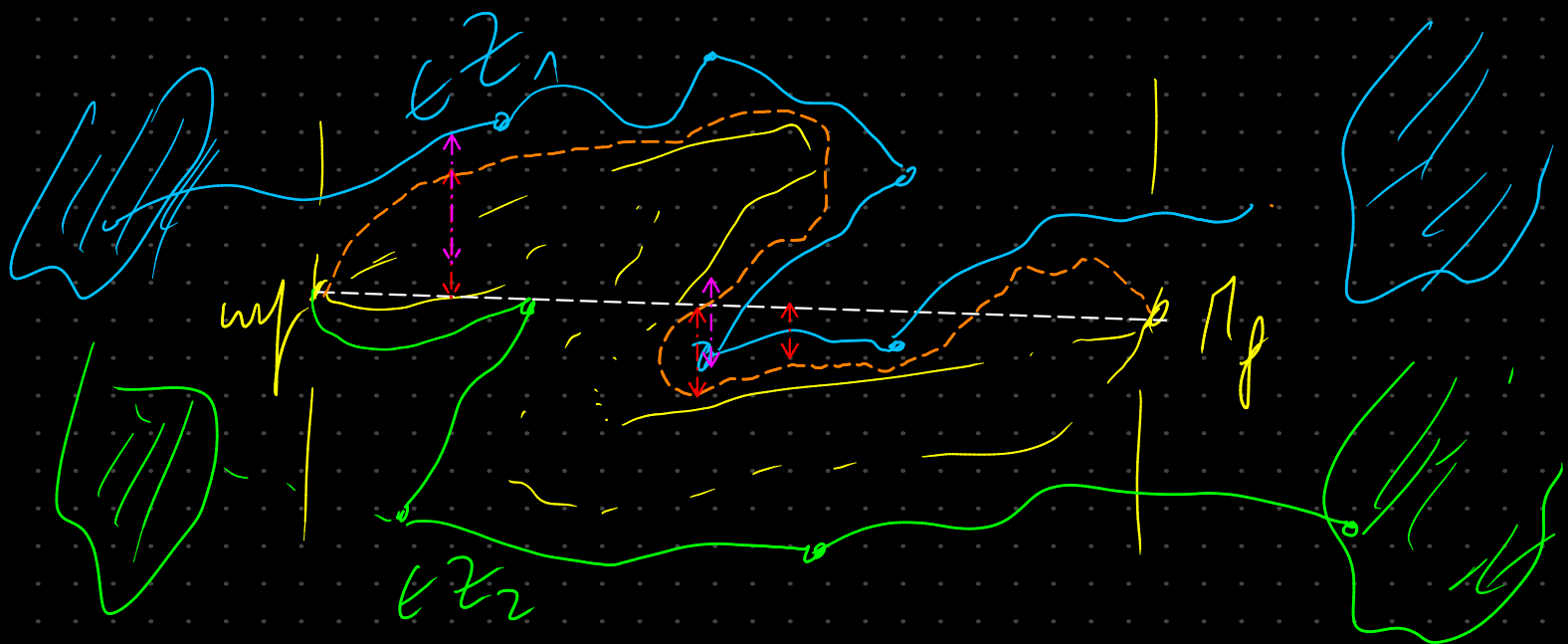
$$\left\{ \begin{array}{l} \forall (x, y') \in Z \quad y' > y \end{array} \right.$$

$$\forall x \in [x(m_f), x(\Pi_f)]$$

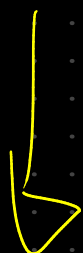
$$\textcircled{\exists y \in \mathbb{R} \text{ s.t. } (x, y) \in B_2}$$

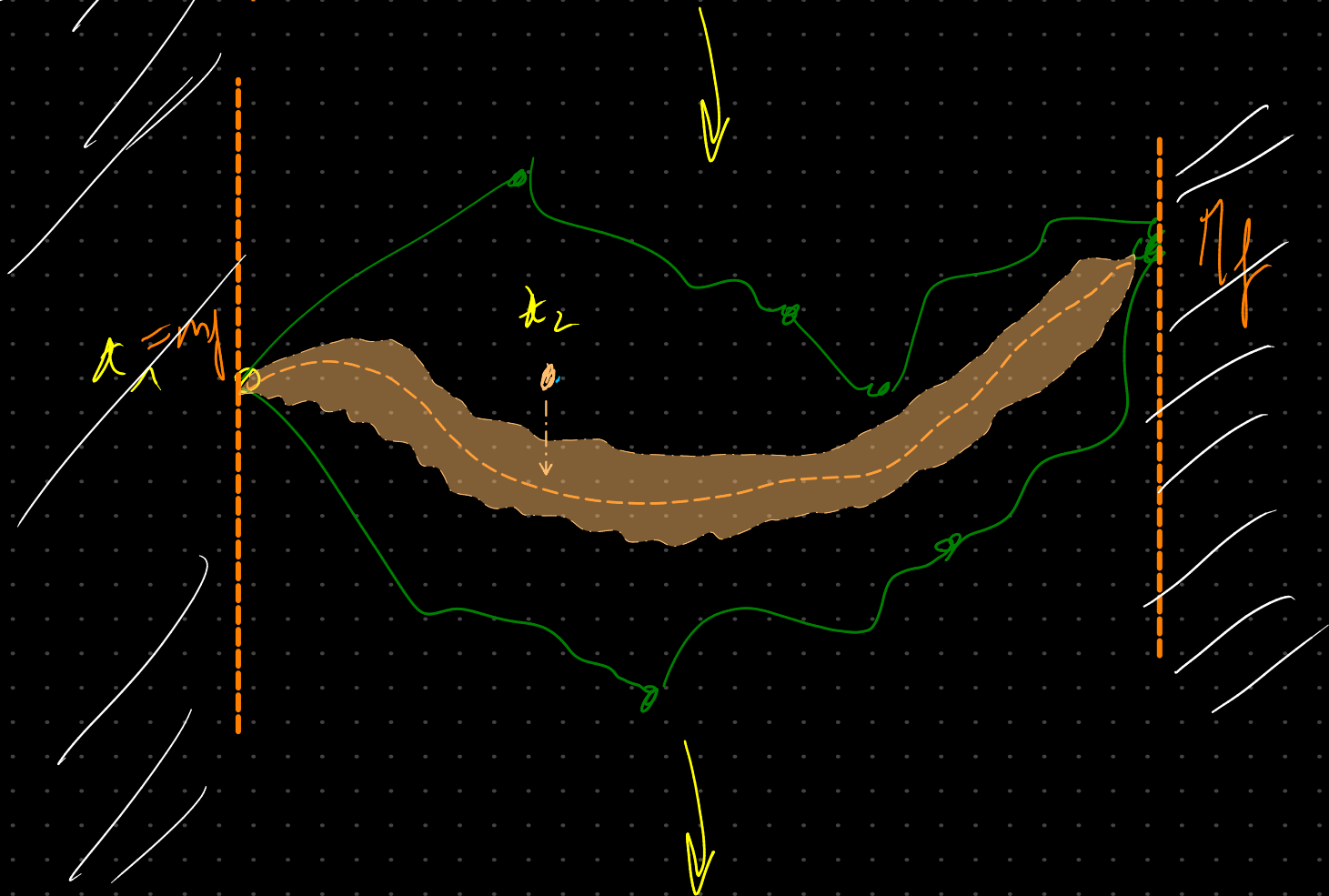
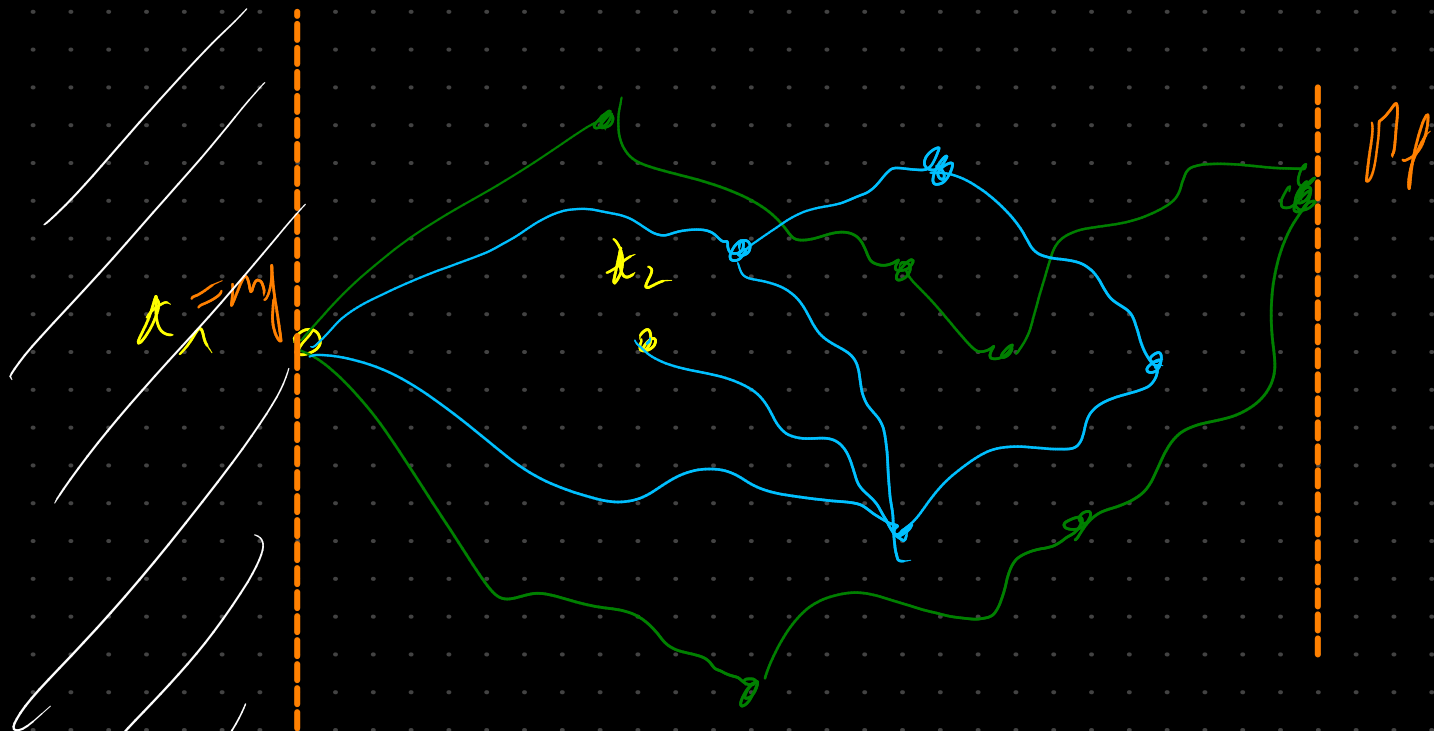
$$\left\{ \begin{array}{l} \forall (x, y') \in Z \quad y > y' \end{array} \right.$$

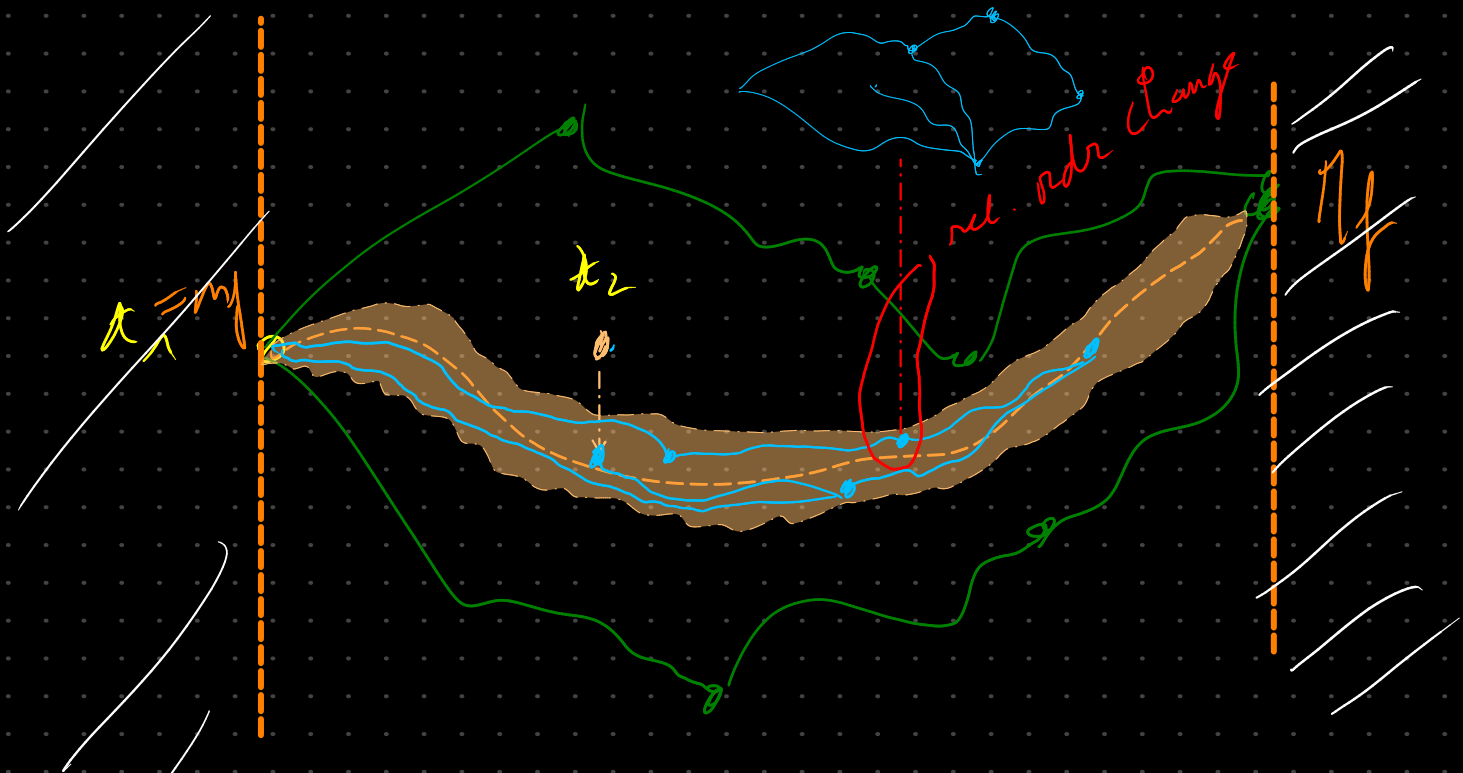




if we go back to case of $h - \{x_2\}$ leading to multiple components in proof of Th (2,1)

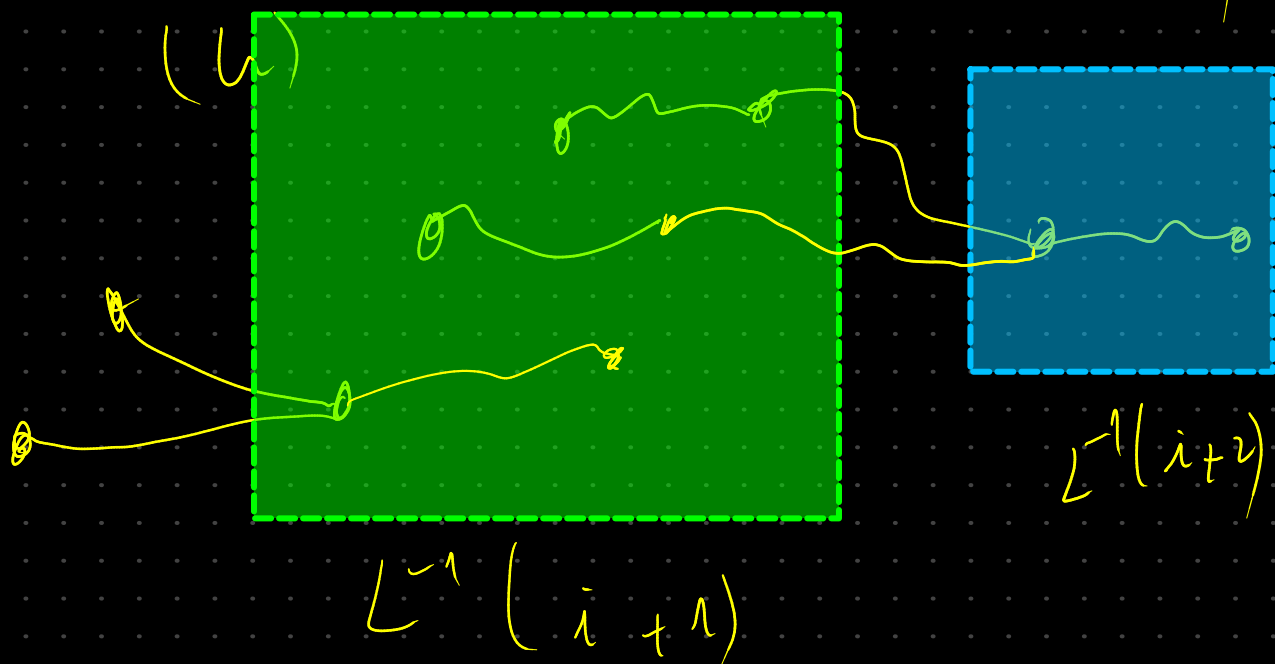


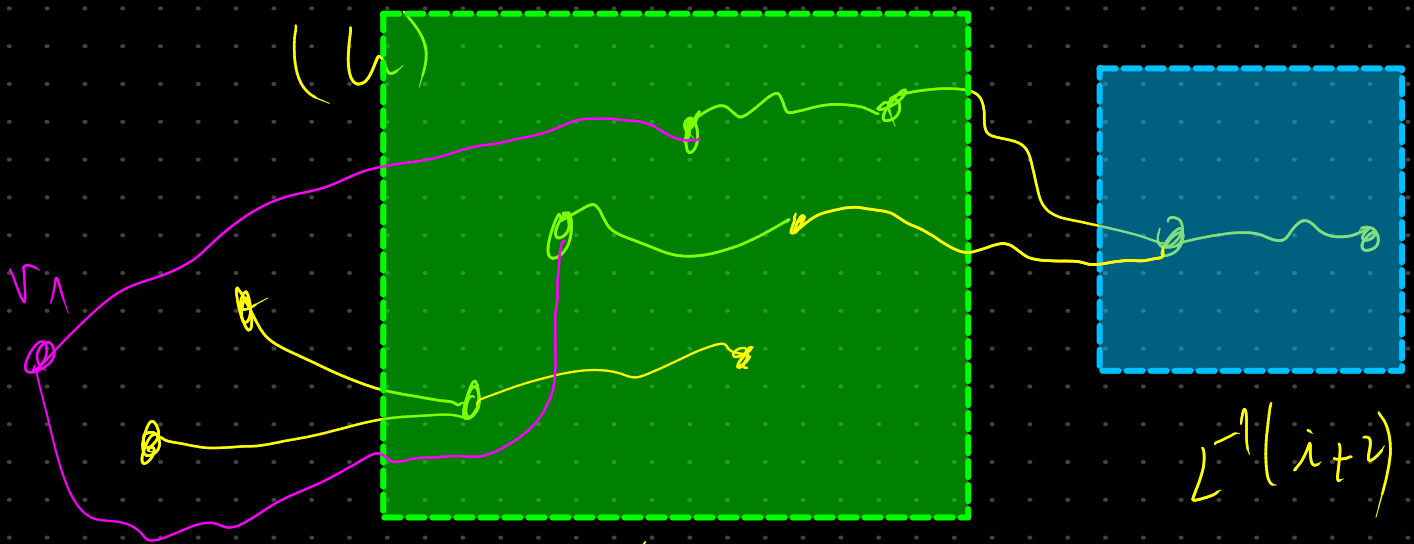




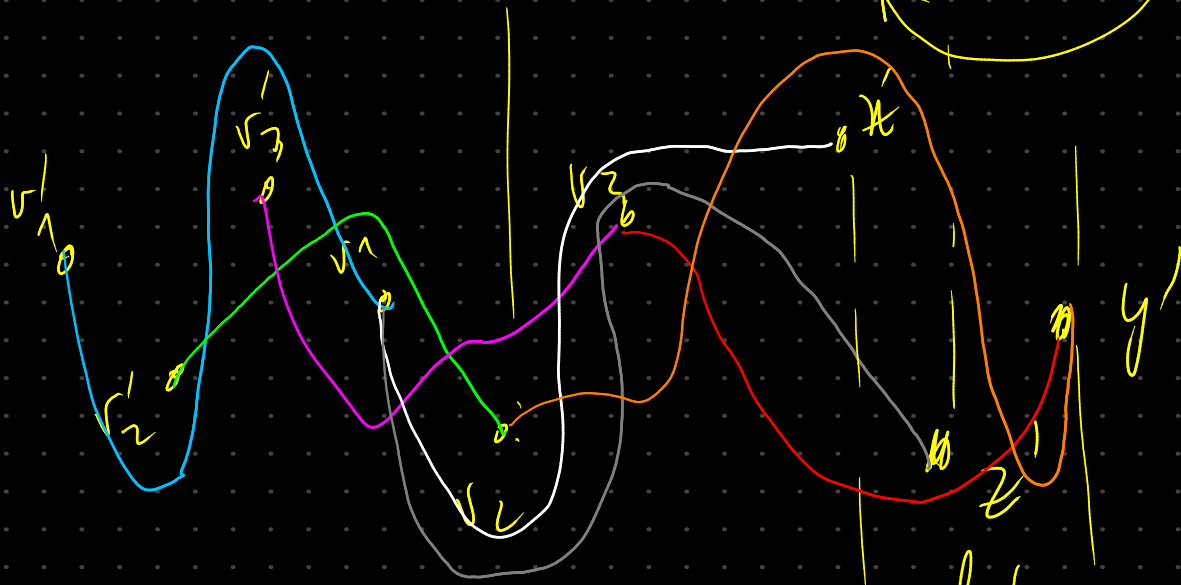
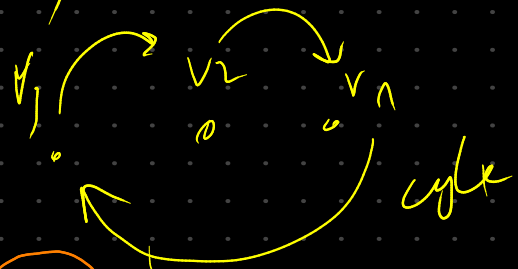
How can I have this curve in h^* ?

1) create a cycle
 from $L^{-1}(i)$ to $L^{-1}(i+2)$
 but first I should draw h (x -monotone
 incl.)





$$L^{-1}(i+1)$$

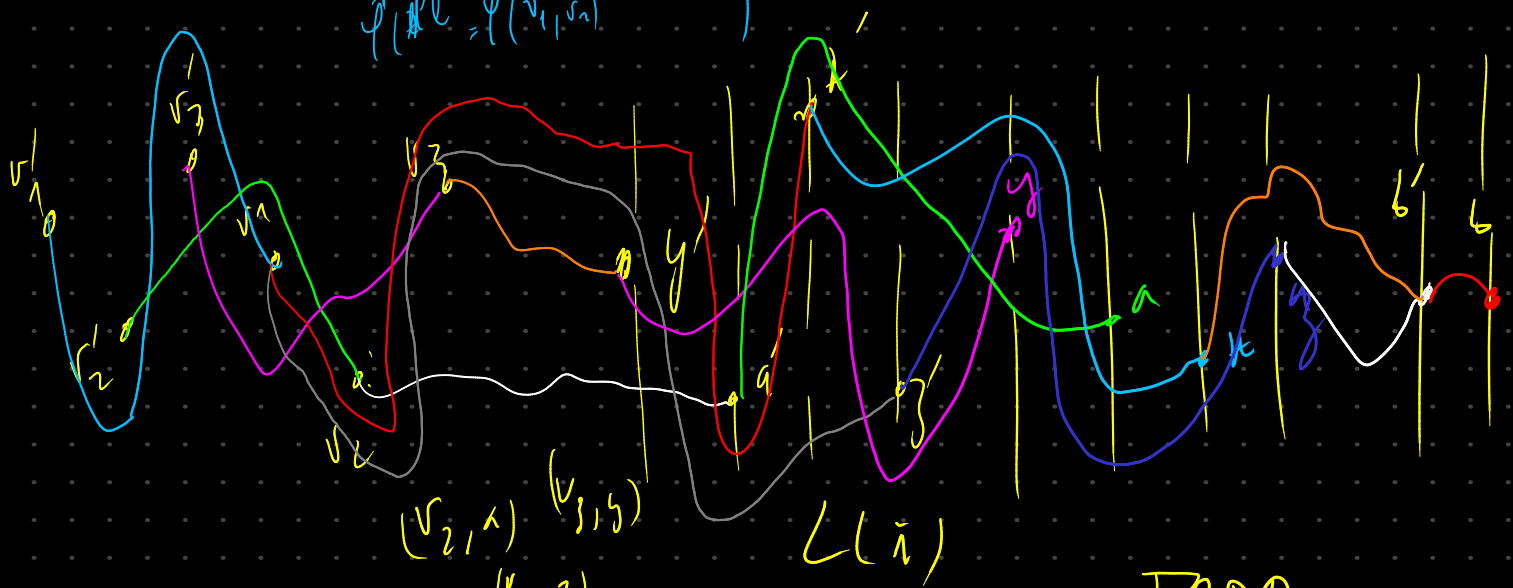


$$\varphi(v_1, v_2) = 0$$

$$\varphi(v_2, v_3) = 0$$

$$\varphi(v_1, v_3) = 1$$

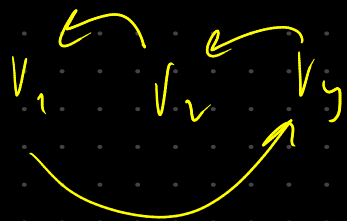
$$\phi(\vec{v}) = \phi(v_1, v_2)$$



$$\phi(v_1, v_2) = 0$$

$$\phi(v_1, v_3) = 1$$

$$\phi(v_2, v_3) = 0$$



$$(v_1, 3)$$

$$\phi(y, a) = 1$$

$$\phi(y, x) = 0$$

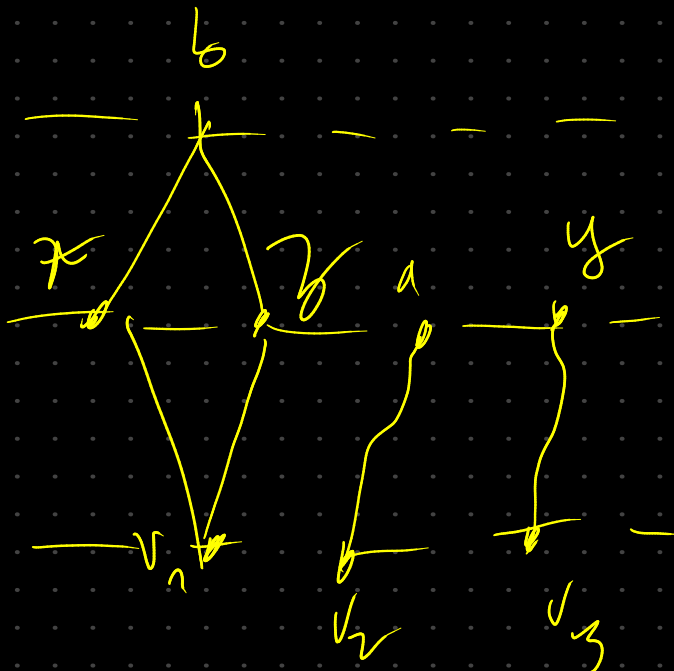
$$\phi(y, z) = 0$$

$$\phi(a, x) = 1$$

$$\phi(x, z) = 1$$

$$\phi(x, z) = 1$$

TODO;
check
even crossings
in
stacked
edges
of $L(\vec{v})$



$$(v_1, a)$$

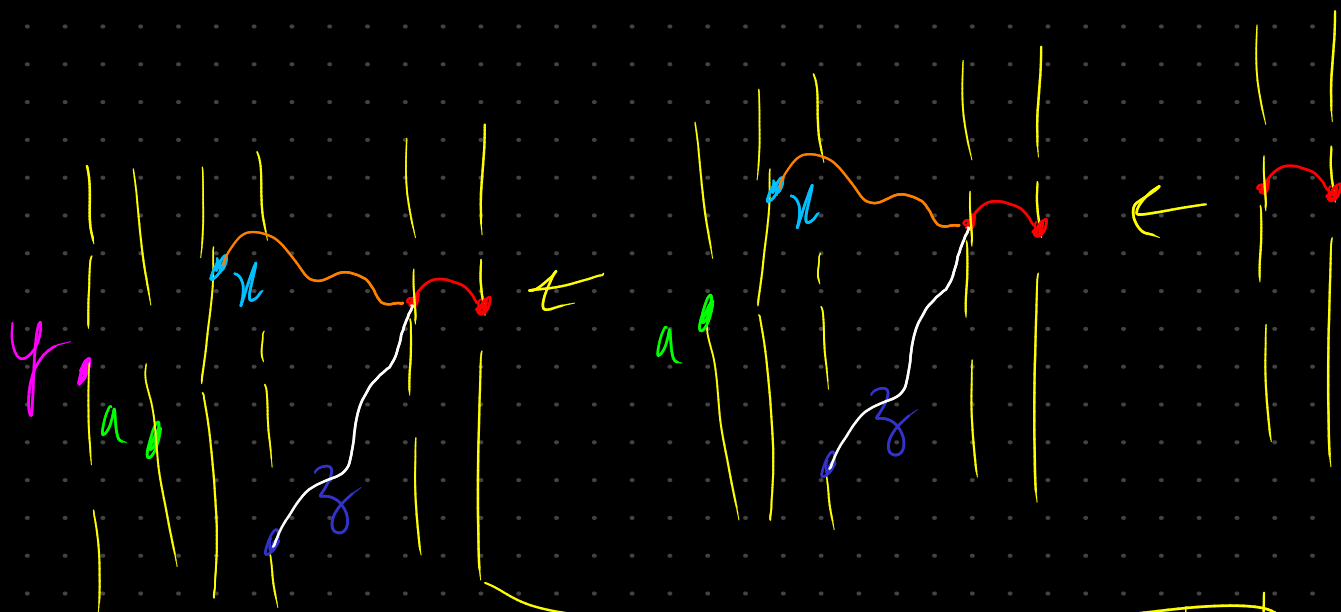
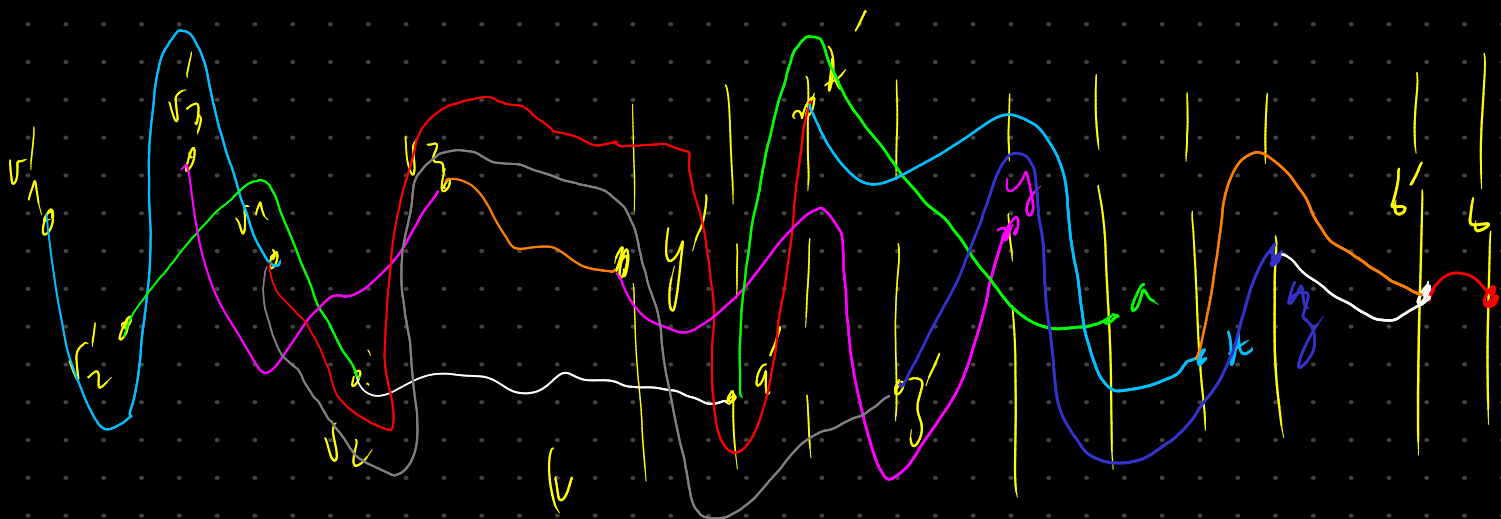
$$(v_3, y)$$

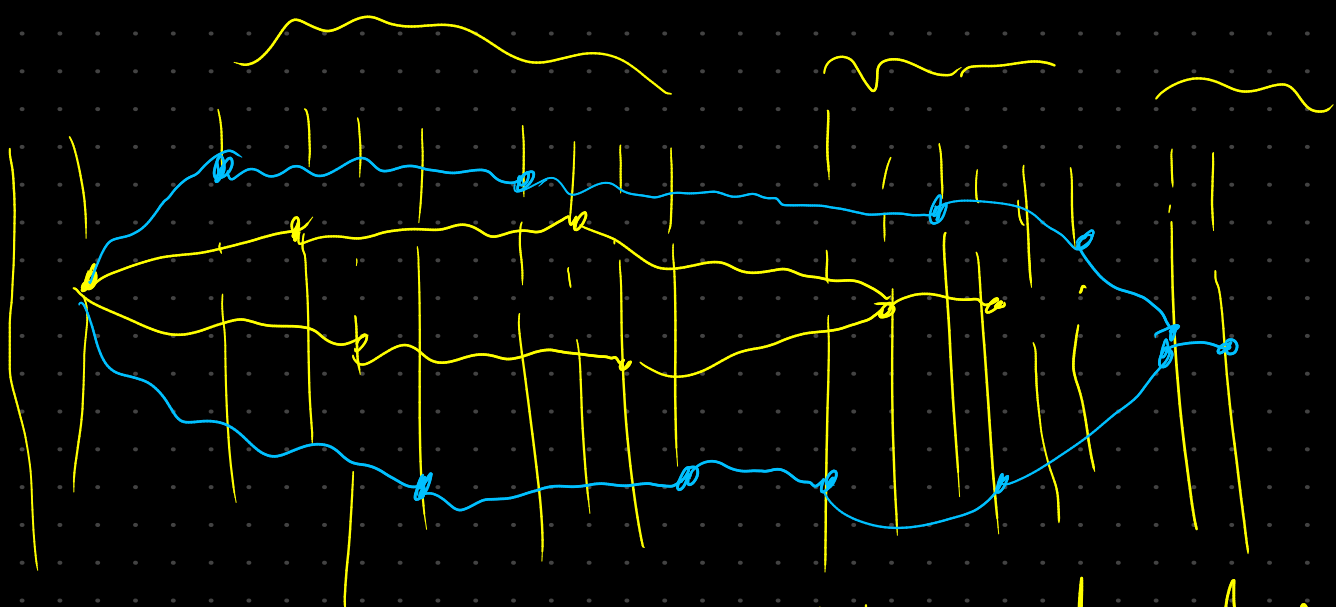
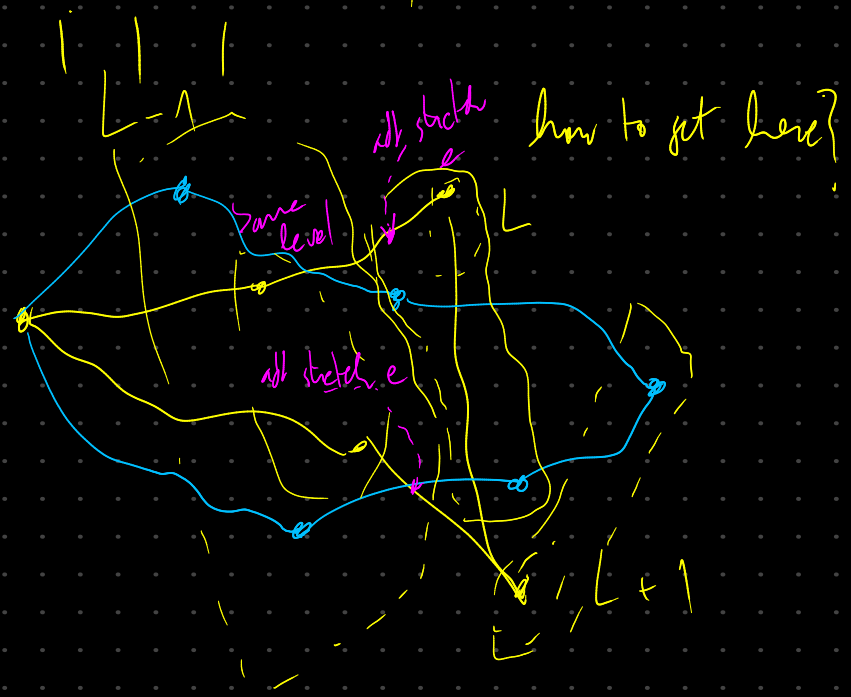
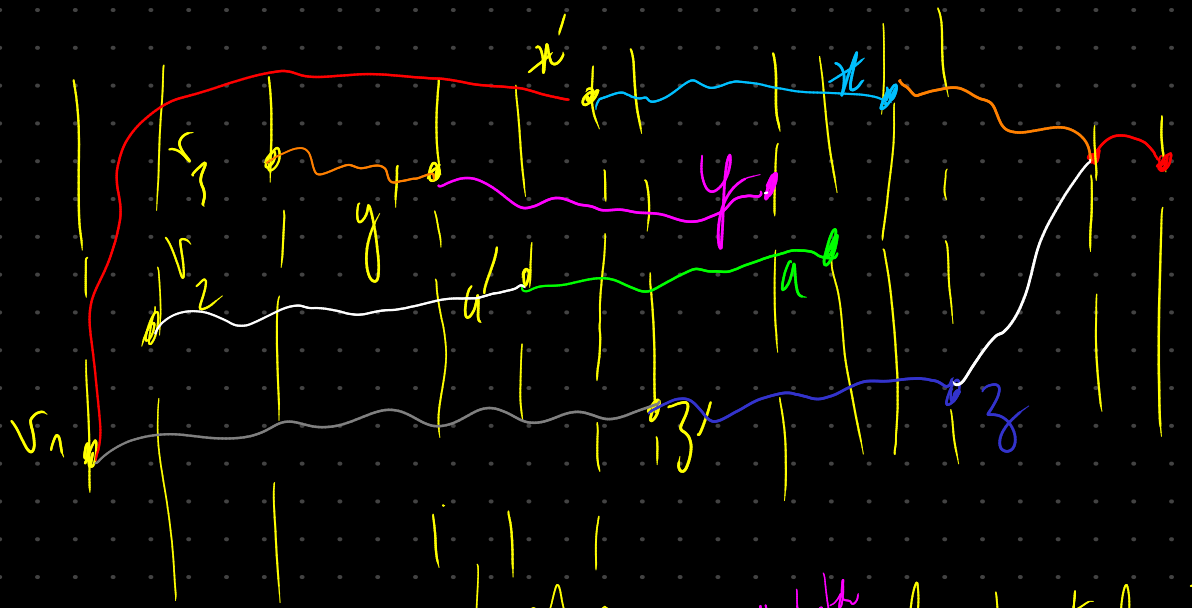
$$(v_1, z)$$

$$(v_1, x)$$

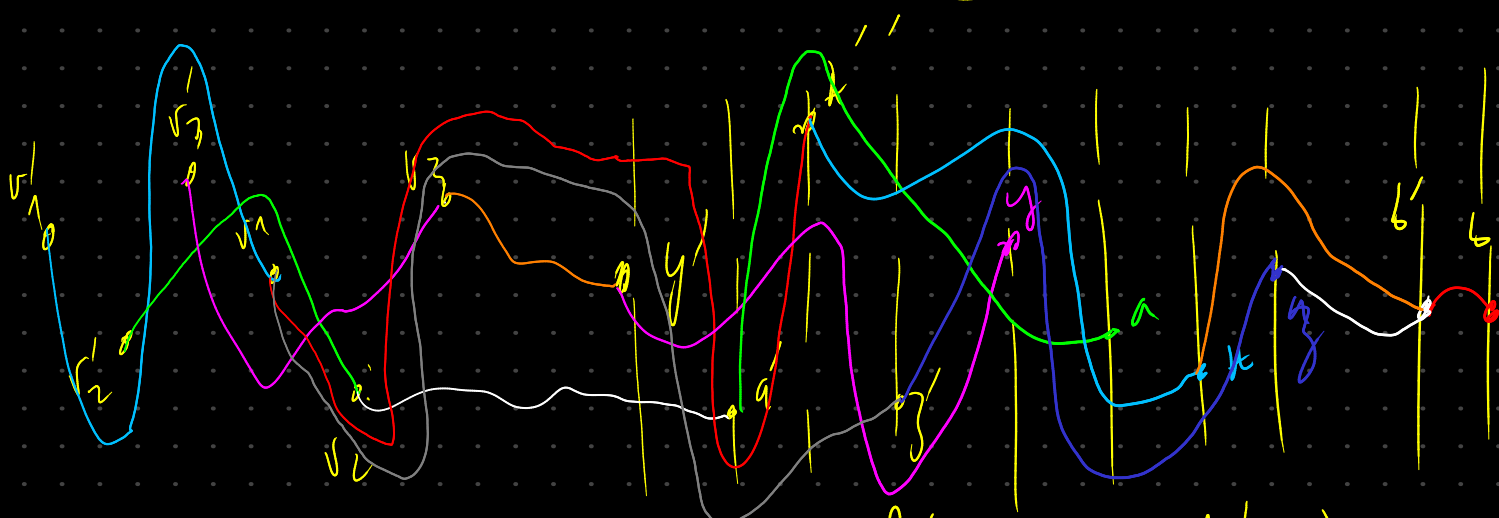
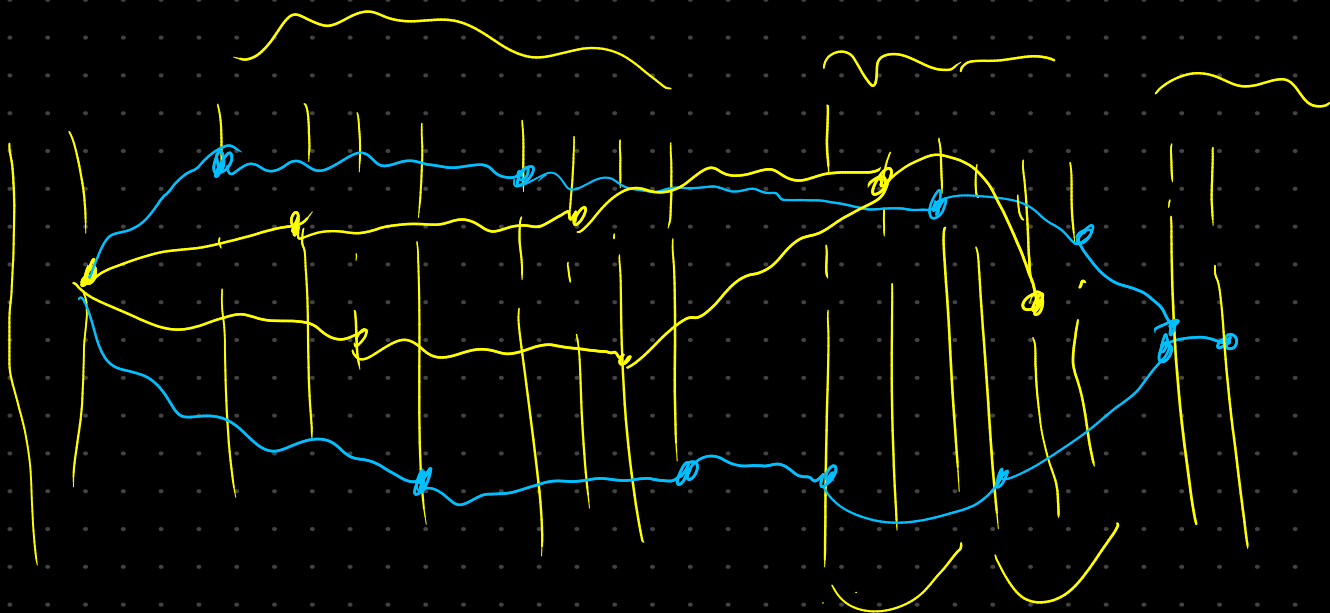
I never managed to find an example for
 $h \sim V(H')$ with $\bigcup_{i=1}^n \mathcal{C}_i$ in proof of Pr 2.1.

Let's check if we actually can get to this case

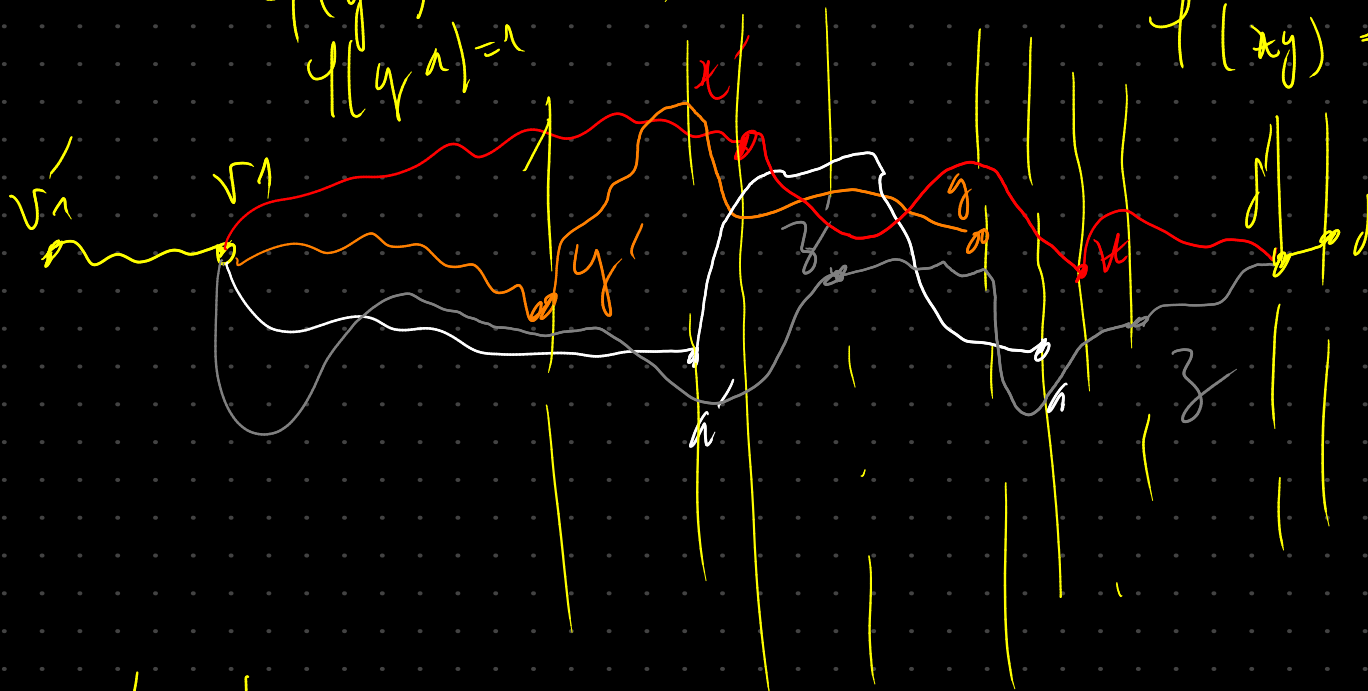




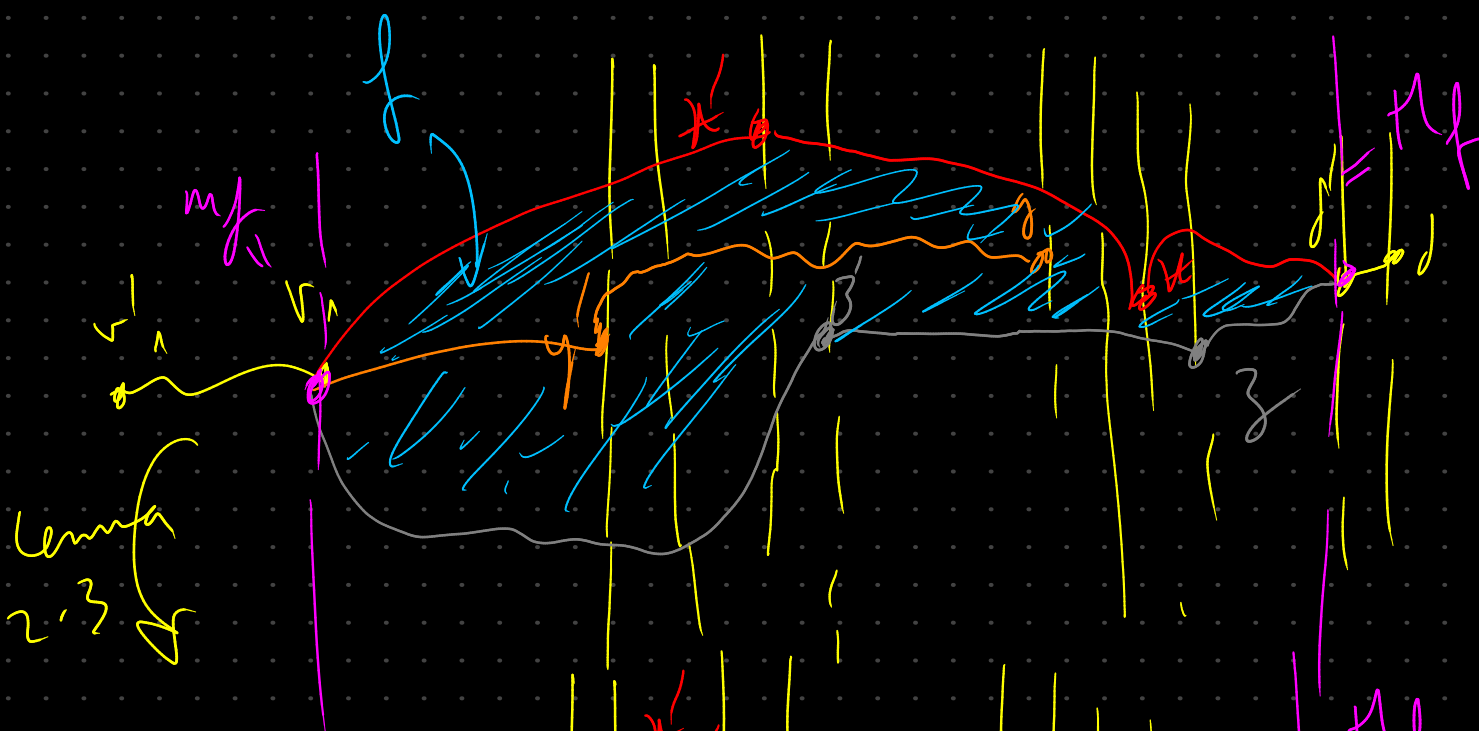
What if we change the relative order of v in yellow?



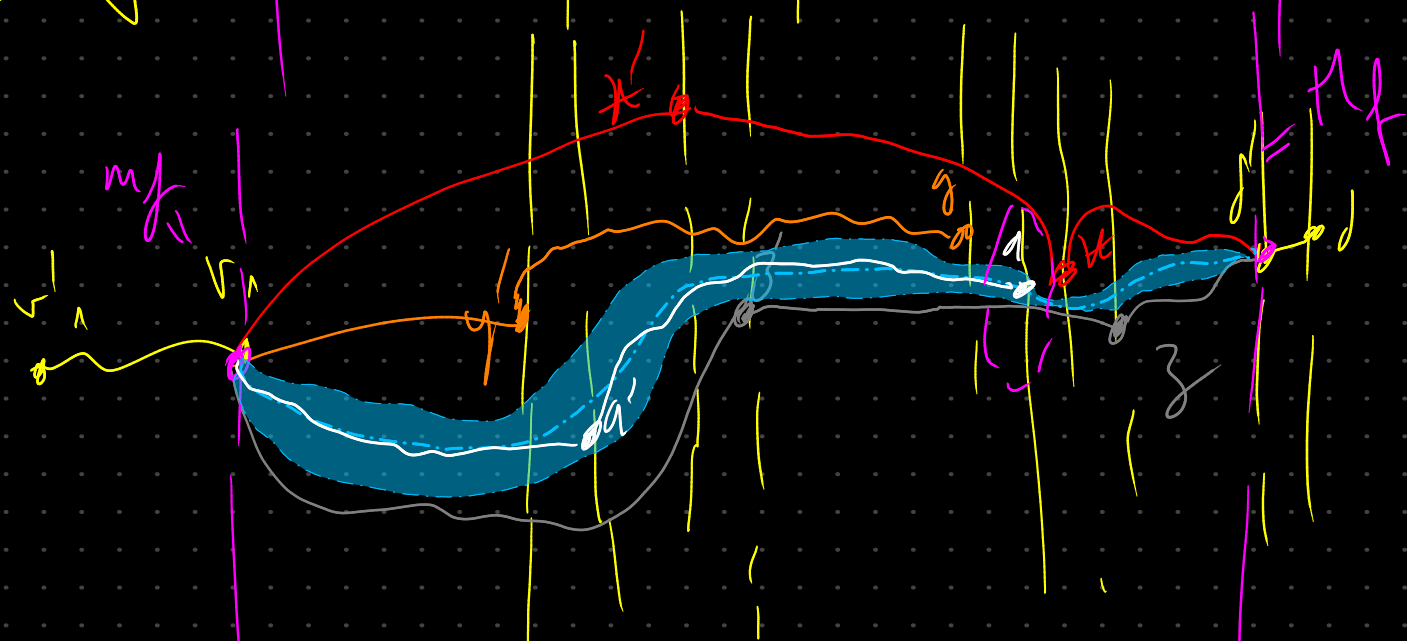
$$\begin{aligned} \psi(yx) &= 0 & \psi(yz) &= 1 & \psi(ax) &= 0 & \psi(az) &= 1 \\ \psi(ay) &= 1 & & & & & \psi(xy) &= 1 \end{aligned}$$



Constructing the x -monad embedding.

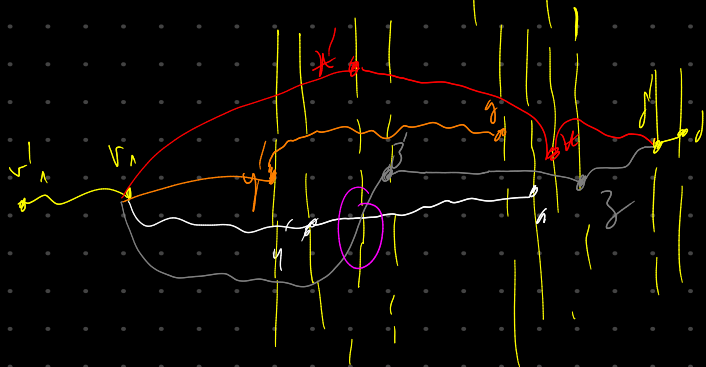
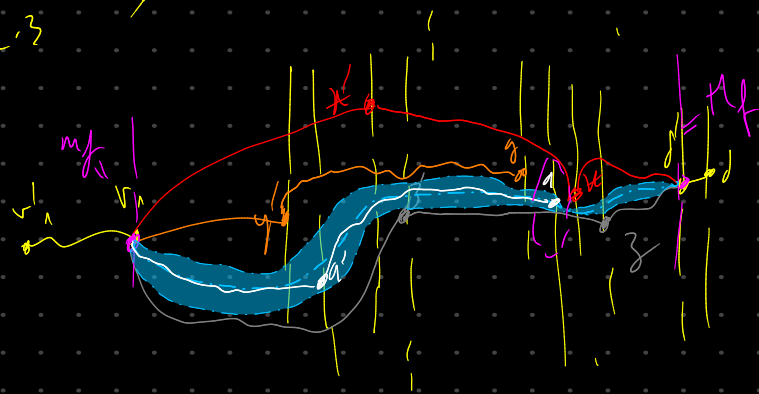
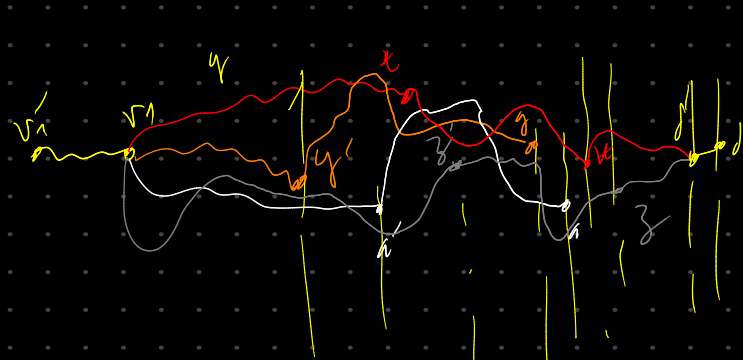


Lemma
2.3

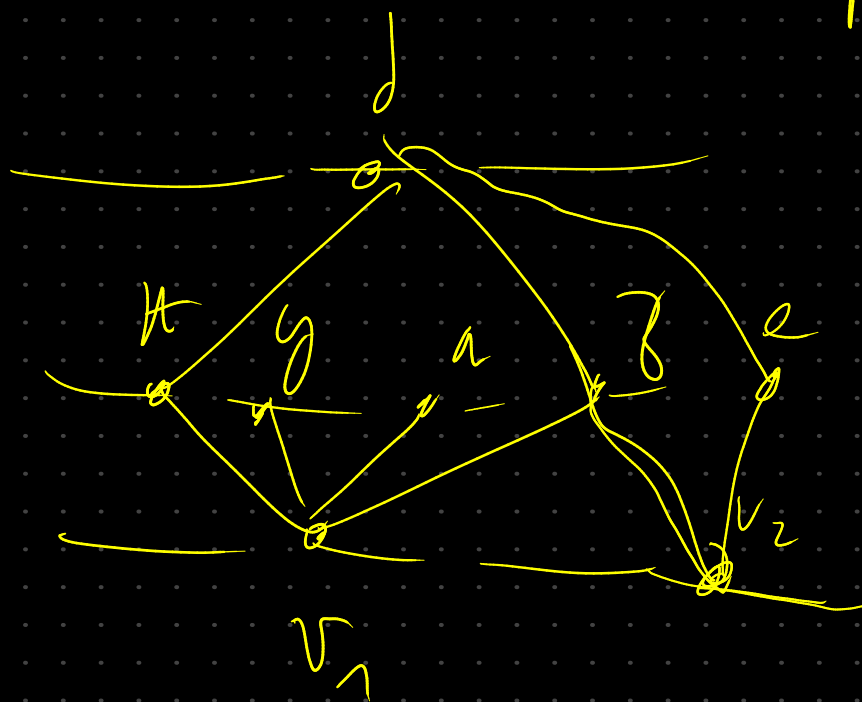


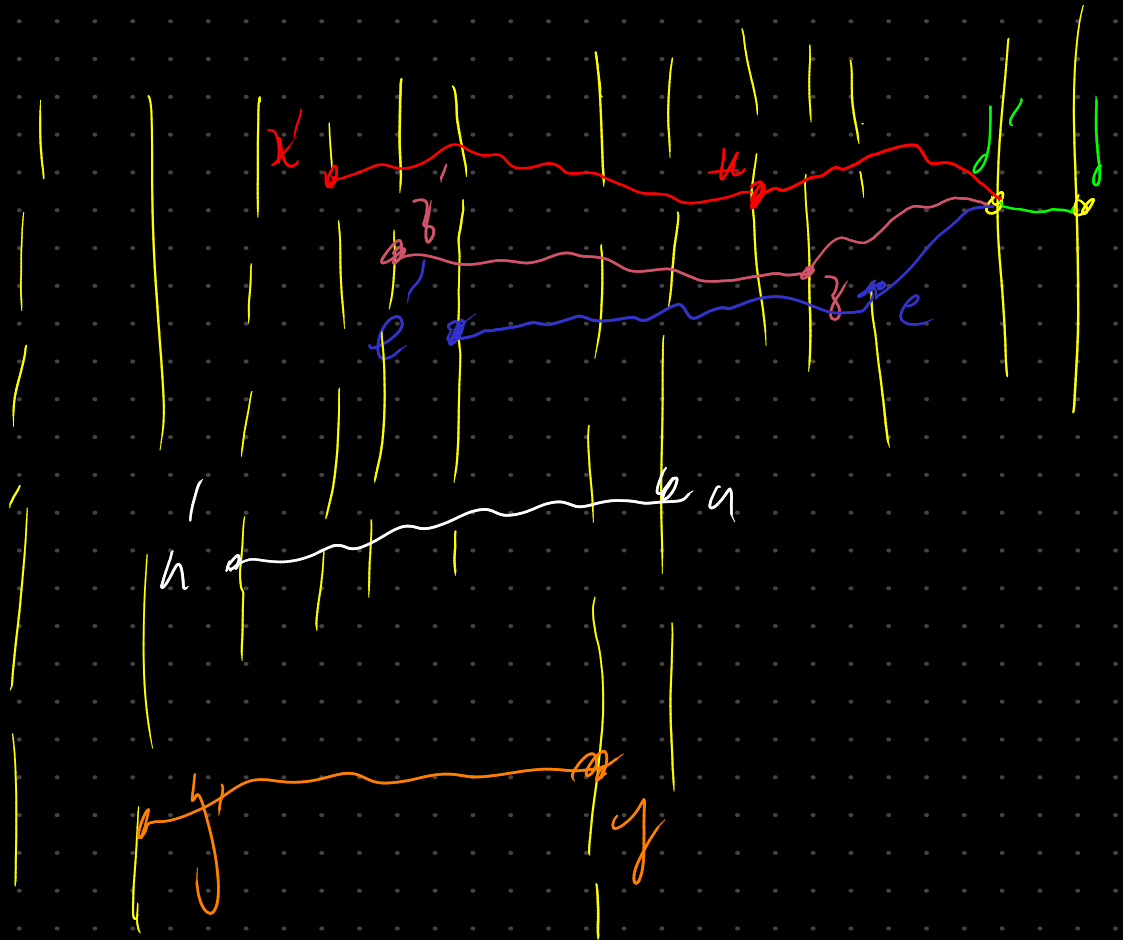
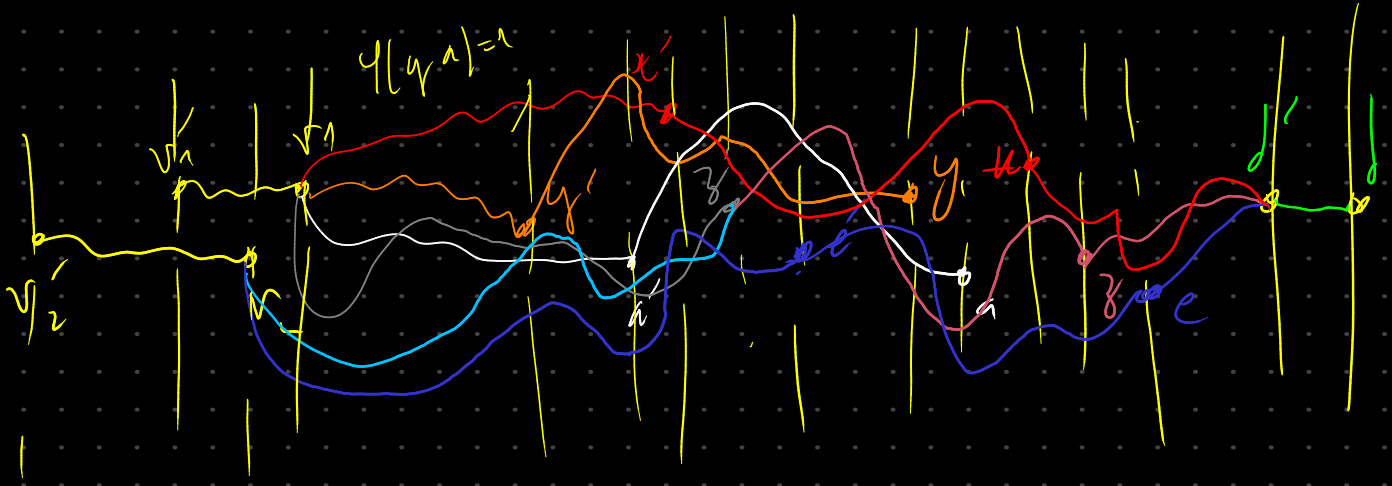
But this only possible if there is no vertex v in the right of v_1 with $L(l(v)) = L(l(v_1))$
 with what does this help us?

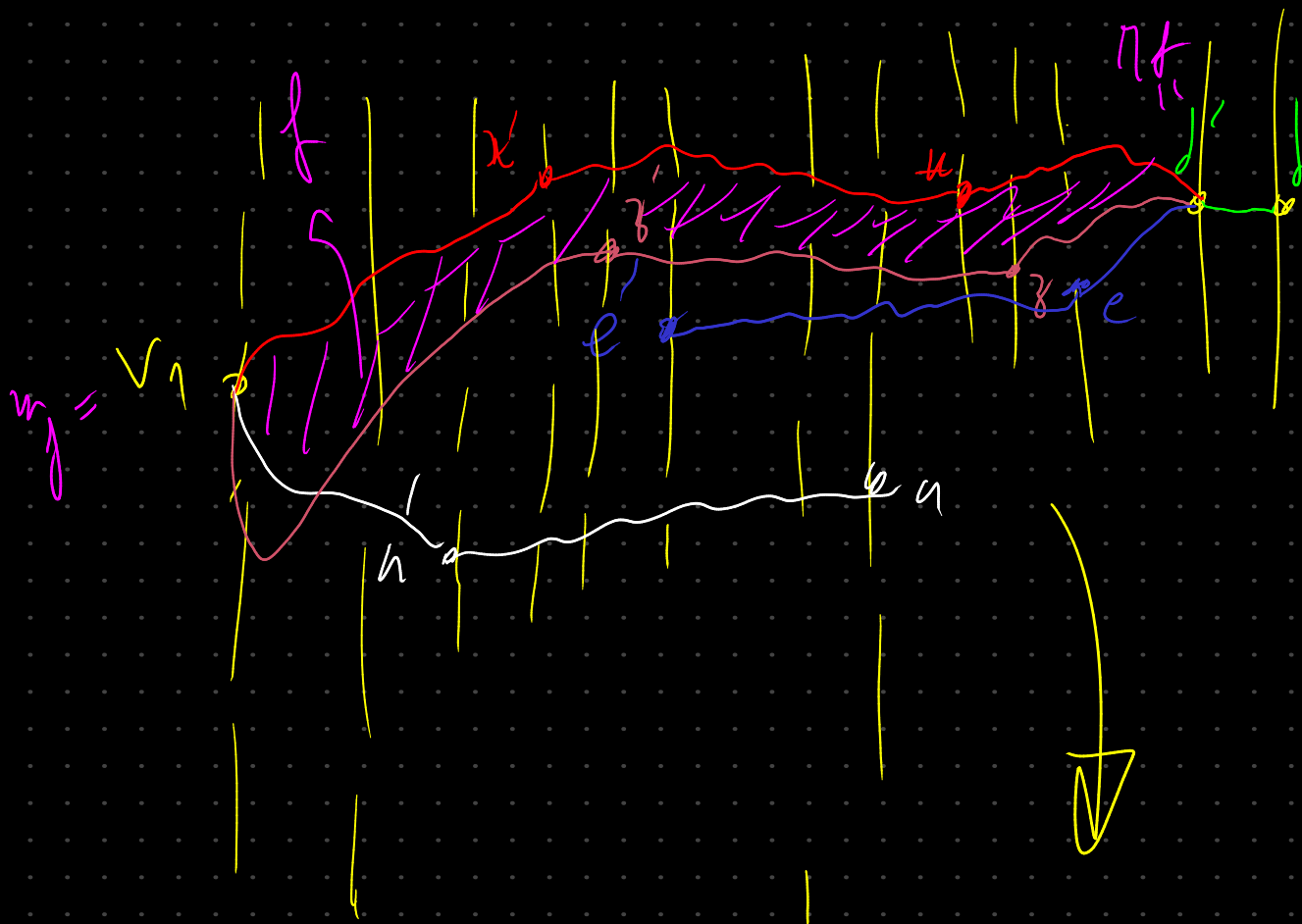
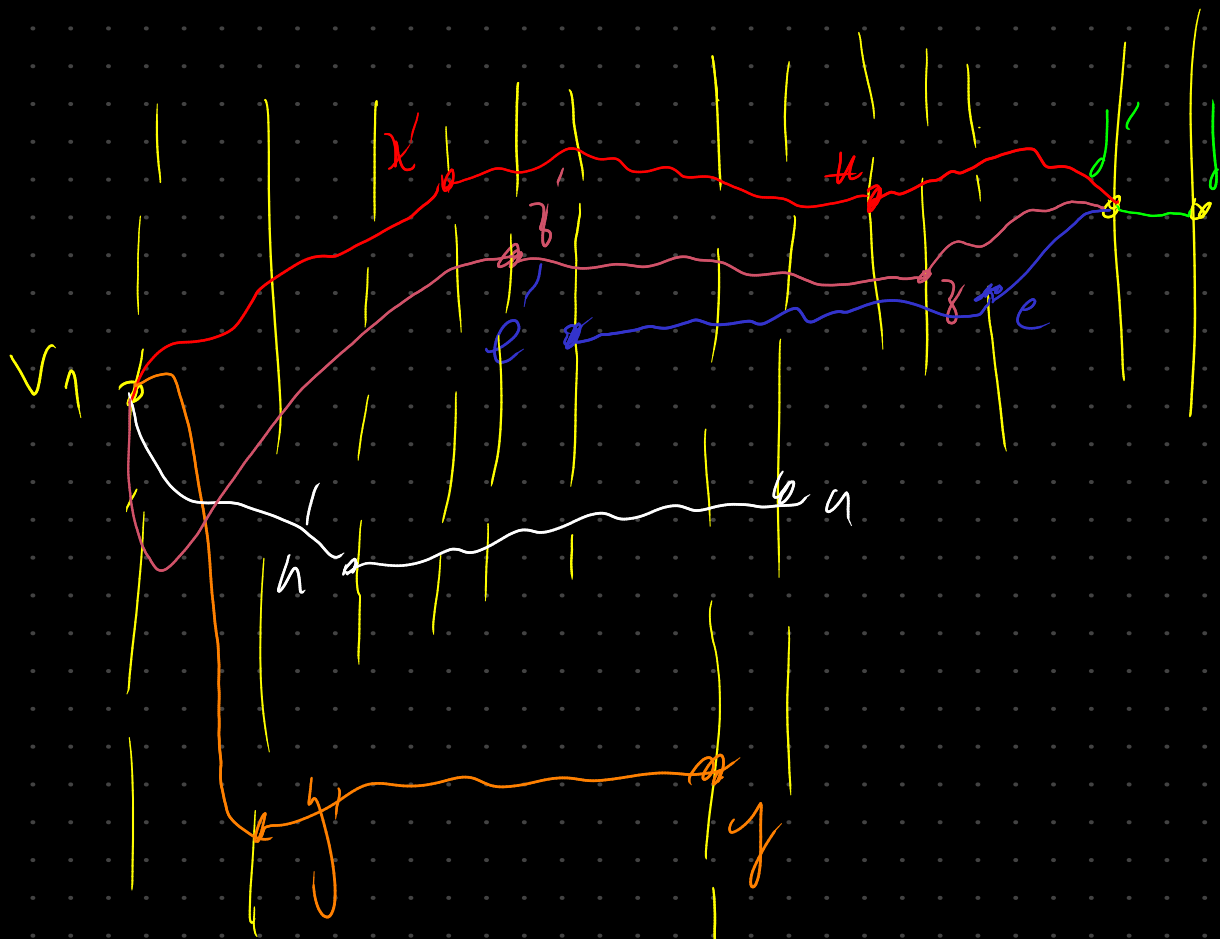
(order doesn't matter tho)



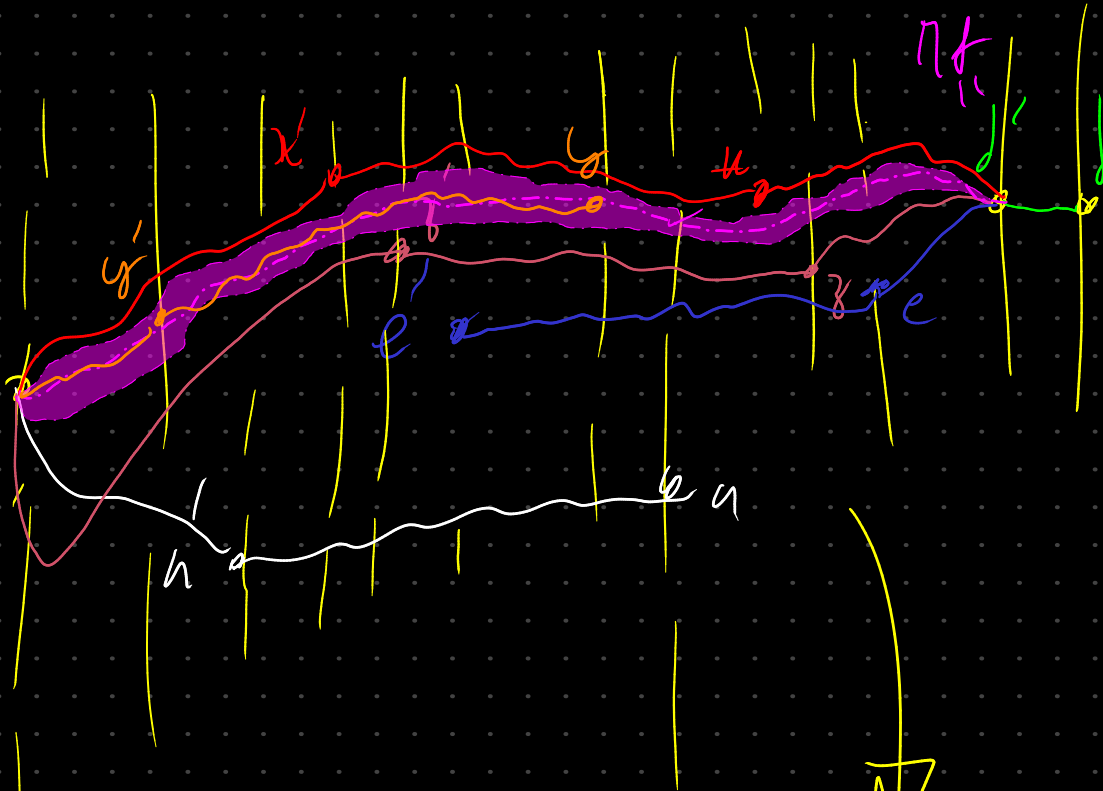
$$\begin{aligned} \varphi(yx) &= 0 & \varphi(ax) &= 0 \\ \varphi(ya) &= 1 & \varphi(az) &= 1 \\ \varphi(yz) &= 1 & \varphi(xz) &= 1 \end{aligned}$$



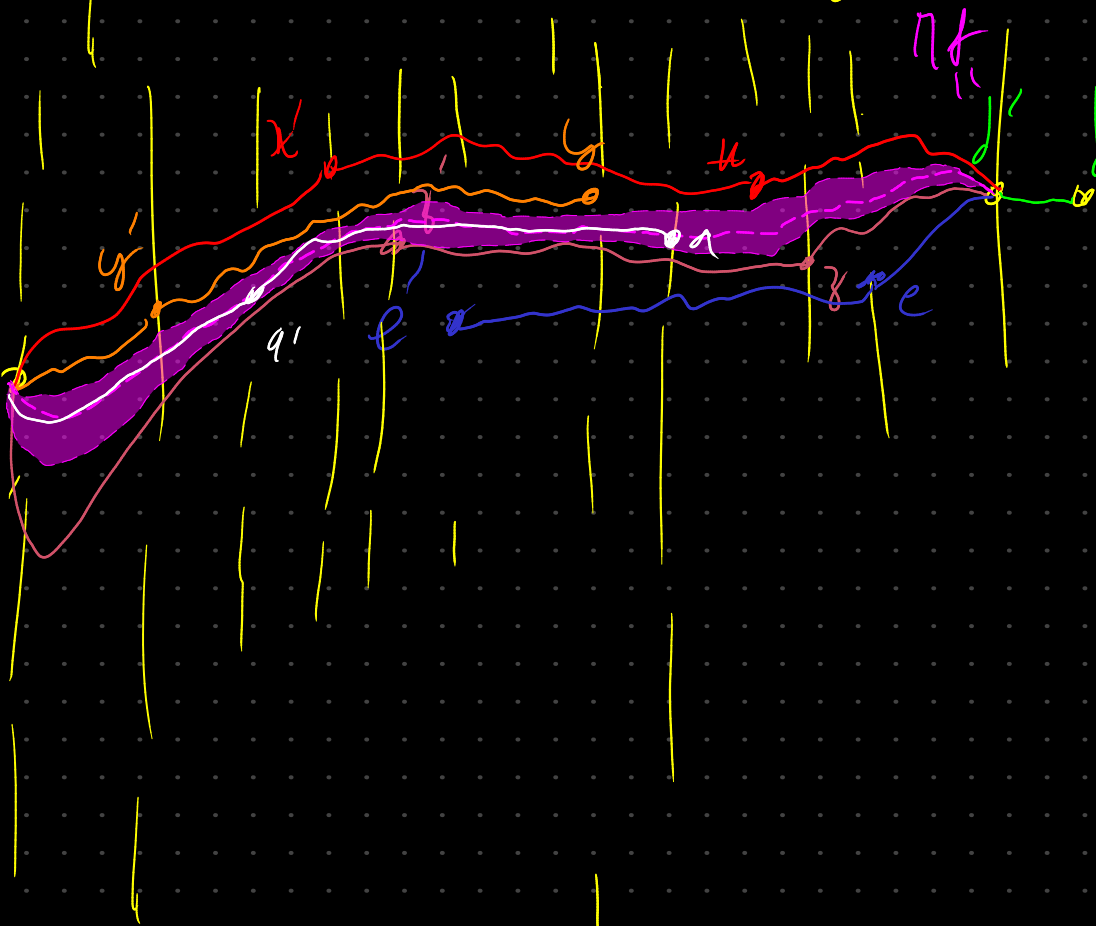


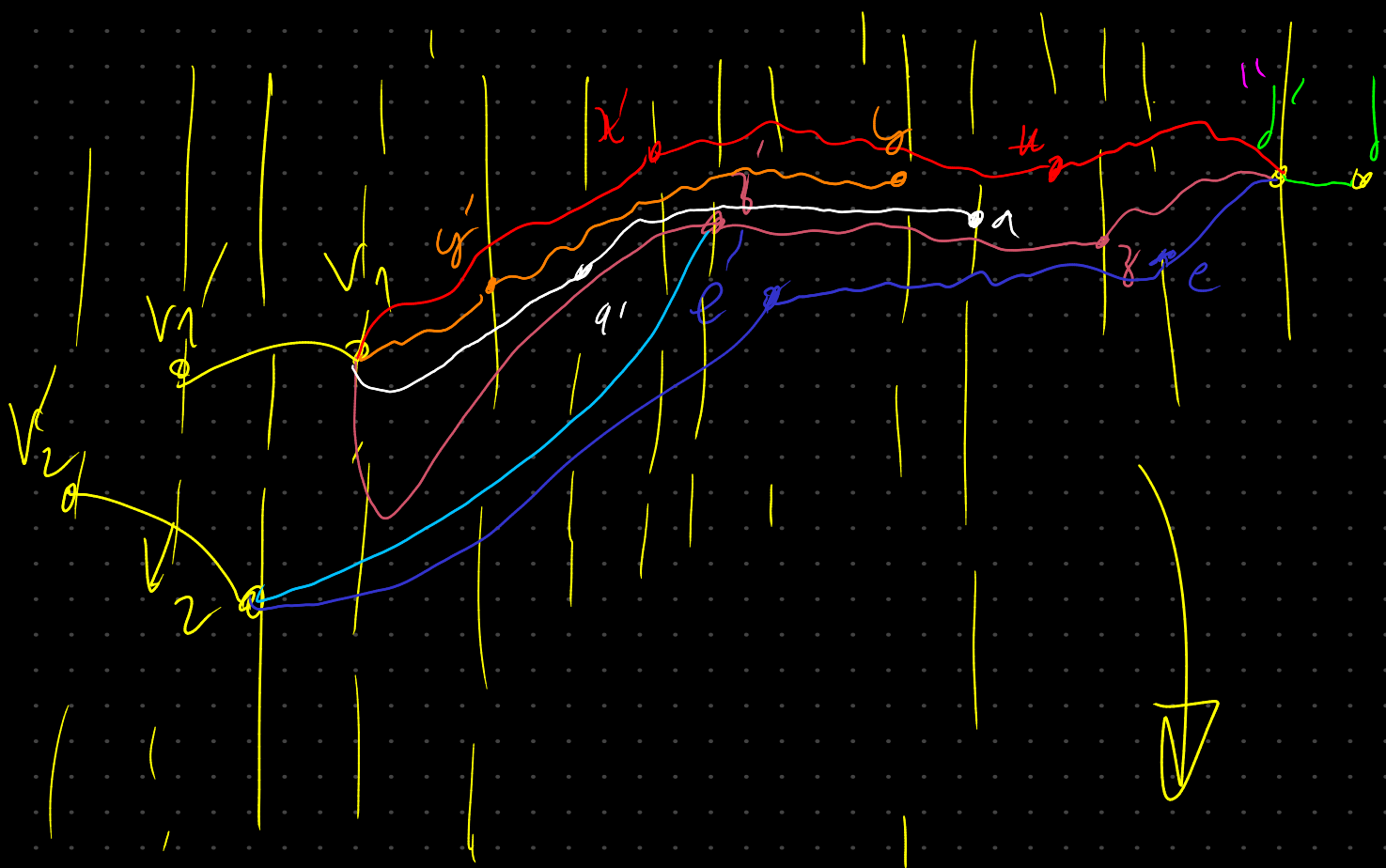


$$w_f = \sqrt{\eta}$$



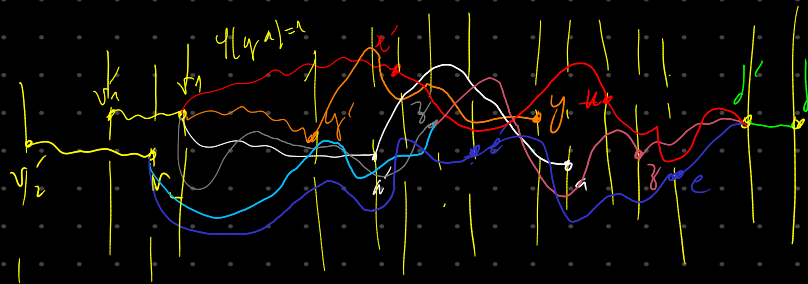
$$w_f = \sqrt{\eta}$$



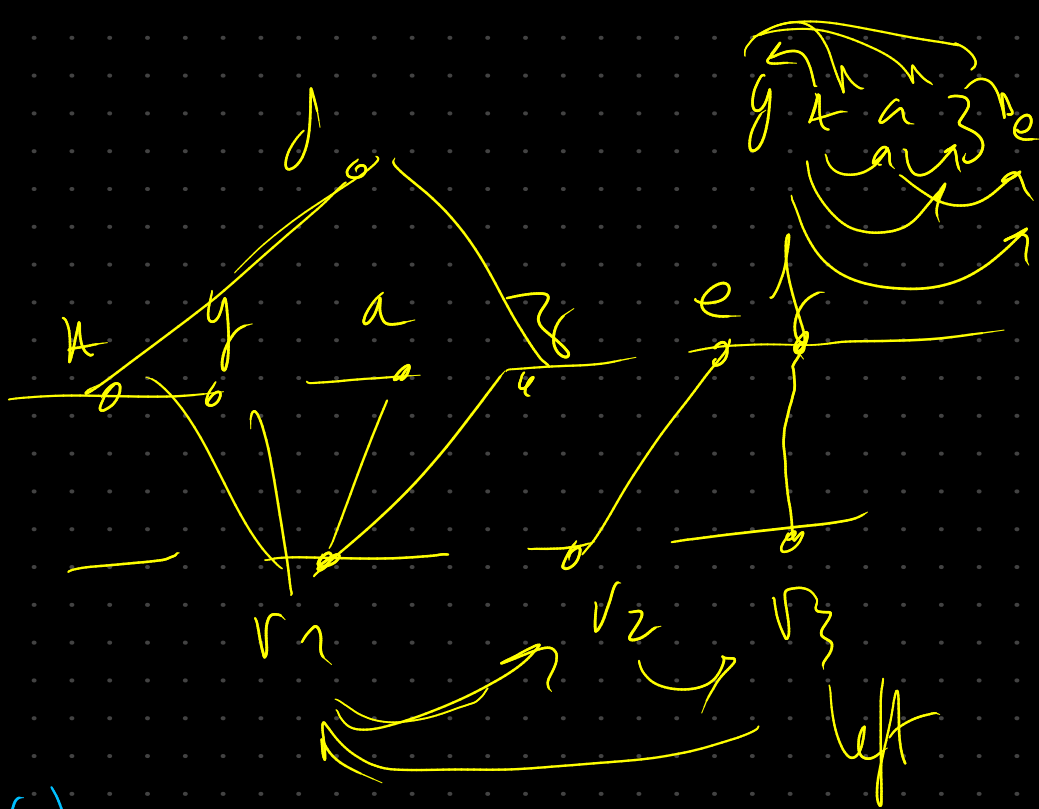


$h - V_2$ is one comp, no issues

$$\begin{array}{ll}
 \varphi(yx) = 0 & \varphi(ax) = 0 \\
 \varphi(ya) = 1 & \varphi(az) = 1 \\
 \varphi(yz) = 1 & \varphi(ae) = 1 \\
 \varphi(ye) = 1 & \varphi(xz) = 1 \\
 & \varphi(xe) = 1 \\
 & \varphi(ze) = 1
 \end{array}$$



$$\begin{array}{ll}
 \varphi(yx) = 0 & \varphi(ax) = 0 \\
 \varphi(ya) = 1 & \varphi(az) = 1 \\
 \varphi(yz) = 1 & \varphi(ae) = 1 \\
 \varphi(ye) = 1 & \varphi(xz) = 1 \\
 & \varphi(xe) = 1 \\
 & \varphi(ze) = 1
 \end{array}$$



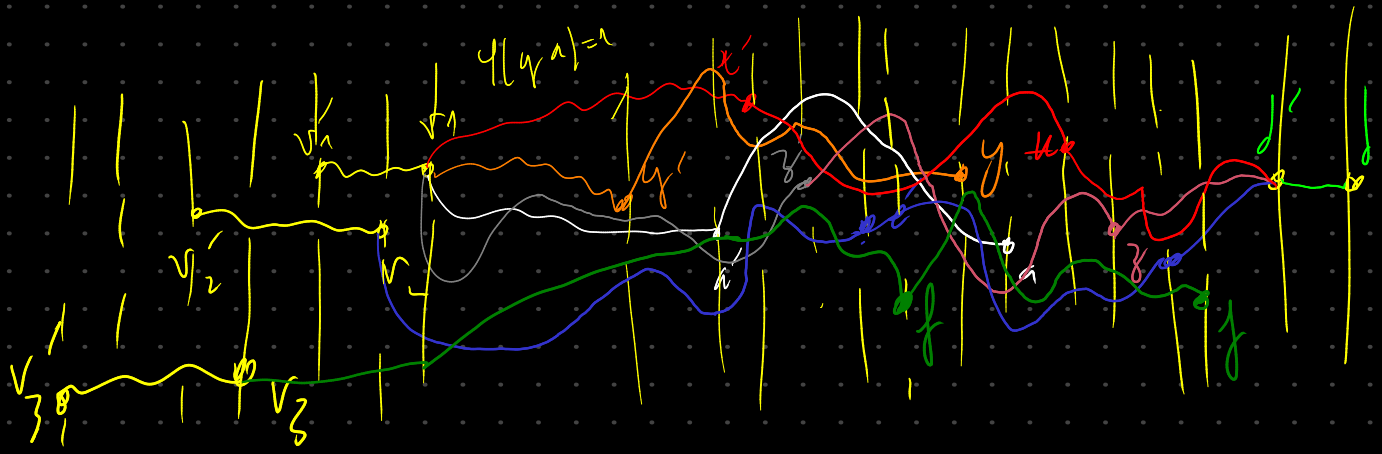
$\ell(v_1 v_2)$

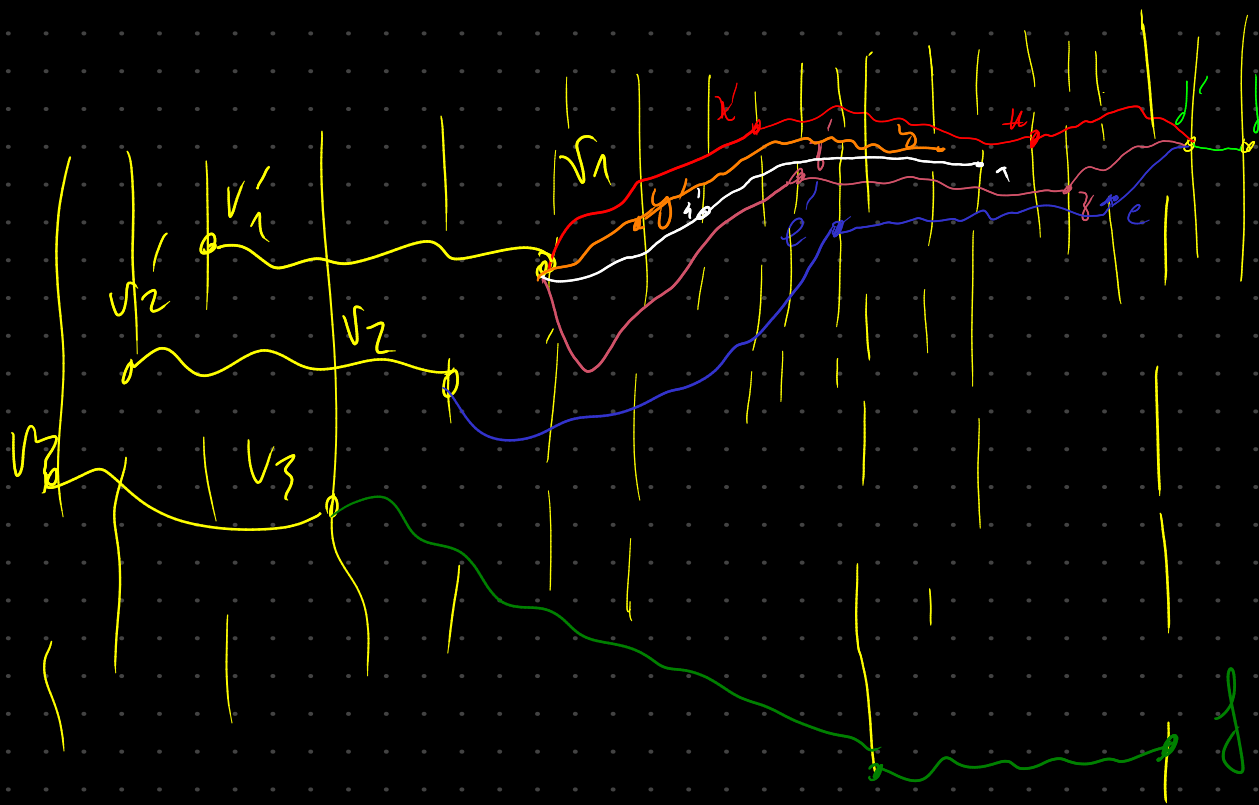
→ $x e x f$
 $y e y d$
 $a e a f$
 $z e z f$

$$\begin{aligned} \ell(v_1 v_2) &= \ell(x e) \\ &= \ell(y e) \\ &= \ell(a e) \\ &= \ell(z e) \end{aligned}$$

$$\ell(v_2 v_3) = \ell(e d)$$

$$\ell(v_1 v_3) = \ell(d \cdot f)$$





$$\begin{aligned} \varphi(v_1 v_2) &= \varphi(xe) \\ &= \varphi(ye) \\ &= \varphi(ze) \\ &= \varphi(e) \\ \varphi(v_2 v_3) &= \varphi(e) \\ \varphi(v_1 v_3) &= \varphi(e) \end{aligned}$$

$$\varphi(xy) = 1$$

